Mixture of Graders: Adaptive Strategy Routing for LLM-Based Mathematical Evaluation

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Abstract

LLM judges are promising for evaluating reasoned mathematical solutions, yet their scores remain prompt-sensitive and unstable. We evaluate 13 grading strategies on Putnam-AXIOM-Grading, a new 500-problem benchmark with 1,000 mathematically annotated candidate solutions, and IMO-GradingBench. The strategies span direct scoring, structured chain-of-thought, and multi-call pipelines such as G-Eval, Chain-of-Verification, and Debate. No fixed strategy dominates, and the strongest single default, comparative prompting, wins only on average. In hindsight, the best strategy per problem differs from the best fixed default in 76% of cases, revealing substantial oracle headroom for adaptive selection. Motivated by this, we train Mixture of Graders, a ModernBERT-large strategy router, using the fixed-strategy score matrix and human labels, then deploy it using only problem and reference features. On Putnam-AXIOM-Grading, MoG improves over the best fixed strategy across all open-weight judge models, with relative MSE reductions up to 27.8% across both Qwen2.5-72B-Math-Instruct and Mistral-Large-Instruct-2411 student responses. It also generalizes across students, with a router trained on Qwen responses outperforming the best fixed default on Mistral responses. The prescription for difficult mathematical grading is not a single best prompt, but a learned mixture over grader strategies.

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1. Introduction

Mathematical benchmarks increasingly shape claims about frontier-model capability: small score differences can decide which systems are treated as state-of-the-art and which methods are selected for further training (Hendrycks et al., 2021; Glazer et al., 2025). But benchmark scores depend not only on the model being evaluated; they also depend on the grader. If reasonable grading protocols assign different partial-credit scores to the same solution, evaluations may partly measure the protocol rather than the model's mathematical ability. This is especially concerning because LLM judges are now central to model comparison and alignment workflows (Zheng et al., 2023; Gu et al., 2024; Chiang et al., 2024; Lambert et al., 2024), while remaining sensitive to prompt format, scoring instructions, and repeated sampling (Liu et al., 2023; Zheng et al., 2023; Shi et al., 2025).

Mathematical grading makes this protocol dependence harder to ignore. Unlike final-answer evaluation, partial-credit grading must assess promising approaches, intermediate lemmas, missing cases, invalid implications, and plausible but flawed derivations. This has motivated process-aware evaluation of reasoning traces (Xia et al., 2024; Lightman et al., 2023; Zheng et al., 2025; Uesato et al., 2022). Yet LLM judges can miss errors in problems they can solve independently (Zheng et al., 2023), overthink or loop when verifying reasoning traces (Khalifa et al., 2025), and remain poorly calibrated when assigning partial credit (Mahdavi et al., 2025). The risk is not merely noisy grading, but systematic mismeasurement from the interaction between model output and grader protocol.

Most judge-design work picks one fixed protocol—a prompt, rubric, pipeline, or evaluator model—and applies it uniformly. We instead ask whether the grading protocol itself should vary by instance. To study this, we evaluate 13 grader protocols on Putnam-AXIOM-Grading, a new 500-problem benchmark for Putnam-style partial-credit grading, and IMO-GradingBench, an existing olympiad-style grading benchmark (Gulati et al., 2025; Luong et al., 2025). The protocol pool spans direct scoring, chain-of-thought variants, comparative prompting, evidence-constrained scoring, and multi-call pipelines such as G-Eval, Chain-of-Verification, Debate, and multi-judge ensembling.

The fixed-strategy sweep shows that no protocol is uniformly best. Comparative prompting—comparing the candidate solution against the reference solution—is the strongest fixed default across our experiments, but this is an average-case result. Unconstrained chain-of-thought often increases variance without improving agreement (Wei et al., 2022; Haldar & Hockenmaier, 2025b; Khalifa et al., 2025). Added pipeline complexity does not reliably help either. G-Eval and Debate are often unstable, while Chain-of-Verification and multi-judge ensembling can be competitive but are not consistently dominant despite higher cost (Liu et al., 2023; Dhuliawala et al., 2024; Chan et al., 2024; Verga et al., 2024).

If the best fixed protocol is only best on average, how much performance is lost by applying it uniformly? We quantify this with an in-hindsight oracle over the 13 strategies, selecting the highest-utility grader per instance under an objective combining human-grade agreement, repeated-run stability, and optional token cost. The oracle is not deployable because it uses human grades, but it reveals substantial adaptive headroom. In our held-out analysis, the oracle-selected strategy differs from the validation-selected best fixed strategy in roughly 76% of cases. This suggests that the right abstraction is not a universal grader prompt, but a selector over grader protocols.

We therefore train **Mixture of Graders**, a MODERNBERT-LARGE router trained from the fixed-strategy score matrix and human labels, but evaluated using only problem and reference features. The router never sees human grades, candidate solutions, or student-model identity at inference; it outputs a distribution over grader strategies and forms an expected routed score. Across two Putnam-AXIOM-Grading student-response sets—QWEN2.5-72B-MATH-INSTRUCT and MISTRAL-LARGE-INSTRUCT-2411—MoG improves over the best fixed strategy for every open-weight judge model, with relative MSE reductions up to 27.8%. In cross-generator transfer, a Qwen-trained router improves over uniform mixing and beats the strongest fixed default on Mistral responses in 94.1% of paired bootstrap resamples.

Contributions.

- **Putnam-AXIOM-Grading.** We release a 500-problem human-graded benchmark for Putnam-style partial-credit evaluation and use it alongside IMO-GradingBench to evaluate 13 grader protocols.
- **Evidence that fixed protocols are insufficient.** Comparative prompting is the strongest fixed default but no fixed strategy dominates across models, benchmarks, and instances; an in-hindsight oracle reveals substantial adaptive headroom over the best fixed choice.
- **Mixture of Graders.** We formulate adaptive grader-strategy routing and train a MODERNBERT-LARGE router that uses the fixed-strategy score matrix and human labels in training but only problem and reference features at inference. We evaluate routing across multiple judge models, held-out Putnam problems, and transfer across candidate-solution generators.

2. Related Work

LLM judges and structured evaluation. LLMs are now widely used as evaluators for open-ended tasks (Dubois et al., 2025; Liu et al., 2023). Strong judges can correlate with human preferences while exhibiting biases such as position and verbosity effects (Zheng et al., 2023), motivating rubric-based, multi-criterion, and structured evaluation protocols. Prior work has explored explicit evaluation steps (Liu et al., 2023), criterion-wise scoring (Hashemi et al., 2024), dynamic rubric generation (Du et al., 2024), locked rubrics (Hong et al., 2026), and fine-grained skill-level protocols (Ye et al., 2024). Another line trains evaluator models directly, including PandaLM (Wang et al., 2023b), Prometheus (Kim et al., 2023), and JudgeLM (Zhu et al., 2023), though fine-tuned judges may behave like task-specific classifiers and generalize poorly (Huang et al., 2025). These works improve individual judges, rubrics, or protocols; we instead ask whether the evaluation protocol itself should be selected adaptively rather than fixed globally.

Chain-of-thought, verification, and multi-step judging. CoT improves solving performance (Wei et al., 2022), especially with self-consistency (Wang et al., 2023a), but its value for judging is less clear. Longer rationales can increase variance and prompt sensitivity (Haldar & Hockenmaier, 2025a), self-correction without external signal is unreliable (Huang et al., 2023), and generated rationales may be unfaithful (Lanham et al., 2023; Turpin et al., 2023). Judge-specific studies likewise find limited CoT gains under clear criteria (Yamauchi et al., 2025) and document instruction sensitivity, looping, and overthinking in thinking judges (Khalifa et al., 2025). Verification and aggregation methods address these concerns by separating drafting from checking, as in CoVe (Dhuliawala et al., 2024) and process-level supervision (Lightman et al., 2024), or by aggregating multiple perspectives through debate and panels (Chan et al., 2024; Verga et al., 2024). We treat these methods not as universally better replacements for direct grading, but as candidate grader behaviors whose usefulness may vary by instance.

Adaptive routing and evaluation protocols. LLM routing systems such as FrugalGPT, Hybrid LLM, RouteLLM, BEST-Route, and RouterBench route among models or cascades to trade off quality, cost, and compute (Chen et al., 2024; Ding et al., 2024; Ong et al., 2024; Ding et al., 2025; Hu et al., 2024). Route-To-Reason routes across models and reasoning strategies (Pan et al., 2026), and Symbolic-MoE applies expert selection to reasoning settings (Chen et al., 2025a). Closer to judging, Auto-Prompt Ensemble augments judges with auxiliary evaluation dimensions, while SEAR uses structured evaluation signals for LLM gateway routing (Li et al., 2025; Zhang et al., 2026). In contrast, we do not route among models, gateways, or expert parameters. For a fixed grading task and judge model, we route among *evaluation protocols*—direct scoring, comparison, verification, debate, ensembles, and retry—with an objective tied to human-grade agreement, repeated-run stability, and token cost.

Mathematical reasoning benchmarks and partial-credit grading. Math evaluation has moved from final-answer accuracy toward process-aware assessment. Benchmarks such as MATH and GSM8K emphasize problem-solving accuracy (Hendrycks et al., 2021; Cobbe et al., 2021), while later work studies reasoning quality, process supervision, and error localization (Lightman et al., 2023; Xia et al., 2024; Zheng et al., 2025). Competition mathematics especially requires calibrated partial credit for correct ideas, missing cases, invalid lemmas, and partially successful proof strategies. Recent grading work shows that LLMs can flag incorrect solutions but remain poorly calibrated when assigning partial credit (Mahdavi et al., 2025). Putnam-AXIOM targets high-difficulty reasoning with partial credit and error detection (Gulati et al., 2025), while IMO-GradingBench provides human-graded olympiad-style solutions (Luong et al., 2025). We build on this process-aware grading setting, but shift the question from which judge or prompt is best on average to whether grader protocols should be selected adaptively for each problem.

3. Methods

3.1. Task and data

We evaluate fixed and adaptive LLM graders on two competition-mathematics grading benchmarks: Putnam-AXIOM-Grading and IMO-GradingBench (Luong et al., 2025). Each grading instance contains a problem statement, a reference solution, a model-generated candidate solution, and a human reference label. We first run a fixed-strategy sweep over grader protocols, then use the resulting strategy–instance score matrix to study and train Mixture-of-Graders routing.

Putnam-AXIOM-Grading. We introduce Putnam-AXIOM-Grading, a 500-problem partial-credit grading benchmark derived from Putnam-AXIOM (Gulati et al., 2025). Each record contains a LaTeX-formatted problem statement, topic and difficulty metadata, a reference solution based on the official Putnam solution, a frozen candidate solution, and rubric-based partial-credit labels. To vary candidate-solution distributions, we generate solutions from two student models in different families, Qwen2.5-72B-Math-Instruct and Mistral-Large-Instruct-2411, yielding 1,000 graded candidate solutions over the 500 problems.

Labels come from mathematically trained annotators rather than official Putnam graders. A primary annotator with competition-mathematics experience graded all 1,000 solutions using the 5-component analytic rubric in Figure 2.[1] Each component is scored in $\{0, 0.5, 1\}$, giving a total 0–5 human reference label. Analytic rubrics can reduce rater variability and make scoring decisions more auditable (Jonsson & Svingby, 2007; Reddy & Andrade, 2010; Eckes, 2023); here, reference solutions also allow component-level judgments to be checked against explicit derivations. A second independent annotator re-graded a stratified 120-record subset covering six difficulty buckets and all 11 topic domains, achieving 90.0% exact agreement on total score, quadratic weighted kappa of 0.983, mean absolute difference of 0.067, and component-level exact agreement of 95%–100%.

IMO-GradingBench. IMO-GradingBench contains 1000 IMO-level student-written solutions with human-assigned scores (Luong et al., 2025). Each example includes the problem statement, a model-authored free-form solution, and one or more human grades on the standard 0–7 olympiad scale. We use the benchmark's partial-credit setting to test whether graders can assess not only final-answer correctness, but also the structure and validity of multi-step reasoning. The judge-facing rubric is shown in Figure 3.[2]

Rubric and output format. Putnam-AXIOM-Grading uses a 0–5 analytic rubric; IMO-GradingBench uses a 0–7 olympiad-style scale. For both benchmarks, all LLM graders receive the benchmark-specific rubric and emit scores in structured JSON enclosed by `<json>` tags, following common LLM-as-a-judge practice (Zheng et al., 2023; Liu et al., 2023). This common output format enables direct comparison across fixed strategies and supplies parsed scores for adaptive-grader supervision.

3.2. Candidate grader pool

We instantiate a candidate grader pool $\mathcal{S}$ spanning single-call prompts and multi-call judging pipelines. Each grader receives the same problem, reference solution, candidate solution, and rubric; graders differ only in how they elicit,

[1] For details on the Putnam rubric, see Section G.

[2] For details on the IMO rubric, see Section G.

structure, or aggregate reasoning before assigning a score. Full prompts appear in Appendix J.1 for Putnam-AXIOM-Grading and Appendix J.2 for IMO-GradingBench.

Single-call graders.

- **No-CoT**: direct JSON scoring with no rationale.
- **Brief-CoT**: 1–2 sentence assessment before scoring.
- **Full-CoT**: free-form step-by-step rationale before scoring, following chain-of-thought prompting (Wei et al., 2022); for Putnam-AXIOM-Grading, the rationale covers each rubric component.
- **Structured-CoT**: fixed summarize $\to$ check key steps $\to$ decide scores template to reduce unstructured reasoning drift.
- **Self-critique**: initial assessment followed by critique and revision, motivated by self-critique/self-correction work (Huang et al., 2023).
- **Bullet-point**: terse criterion-wise justifications tied directly to rubric dimensions.
- **Comparative**: explicit comparison between the candidate and reference solutions, motivated by evidence that comparative judgments can be more reliable than absolute scoring (Zheng et al., 2023).
- **Quote-forcing**: every evaluative claim must cite a direct quote from the model output, constraining hallucinated critiques in unconstrained rationales (Turpin et al., 2023; Chen et al., 2025b).

Multi-call graders.

- **G-Eval** (Liu et al., 2023): generates an instance-specific checklist of atomic evaluation steps, then scores with explicit pass/fail reasoning.
- **Chain-of-Verification (CoVe)** (Dhuliawala et al., 2024): drafts an evaluation, generates verification questions, answers them using the original instance, and finalizes scores conditioned on the verification results.
- **Multi-judge ensemble**: three persona judges—pedantic, holistic, and teacherly—grade independently with free-form CoT; a chair judge synthesizes their judgments into a final score (Verga et al., 2024; Kalra & Tang, 2025).
- **Debate**: a pro-judge argues for maximal credit, a con-judge argues for minimal credit, and an arbiter assigns the final score, following multi-agent debate evaluation ideas (Chan et al., 2024).
- **Overseer-guided retry**: a judge produces a full-CoT evaluation, an overseer audits it for rubric-level errors, and the judge is re-sampled for up to five attempts if errors are flagged; we select the first overseer-approved output or the attempt with the fewest flagged fields (Lightman et al., 2024; Lanham et al., 2023; Chen et al., 2025b).

3.3. Fixed-strategy evaluation protocol

We evaluate each candidate grader as a fixed protocol applied uniformly across instances. Unless otherwise specified, we use four open-weight judge models—QWEN2.5-32B-INSTRUCT, DEEPSEEK-R1-DISTILL-QWEN-32B, QWEN2.5-72B-INSTRUCT, and META-LLAMA-3.1-70B-INSTRUCT—served with batched inference via `vLLM` (Kwon et al., 2023). For every benchmark, candidate-solution set, judge model, and grader protocol, we run five independent trials with seeds 43–47, temperature 0.7, top_p $= 1$, maximum output length 2048, and stop sequences at JSON closing tags. We use stochastic decoding intentionally to measure protocol instability under repeated judge sampling (Tan et al., 2024; Thakur et al., 2024; Haldar & Hockenmaier, 2025a). Overseer-guided retry permits up to five retries, with overseer responses capped at 800 tokens.

For each strategy–instance–run cell, we extract the final numeric score by reading the structured JSON field (`score` or `score_total`); if that fails, a deterministic regex fallback recovers an explicitly printed final score. This avoids treating minor formatting deviations as grading failures while preserving structured-output comparability, following common LLM-as-a-judge practice (Zheng et al., 2023; Liu et al., 2023). We retain the extracted score, repeated-run scores, token usage, and extraction status for every cell.

We evaluate fixed strategies along two axes. *Agreement* is mean squared error (MSE) against the human reference label, using each strategy's mean extracted score across repeated runs; per-benchmark MSE is on the native score scale, cross-benchmark averages normalize by benchmark maximum. *Stability* is the per-instance variance of total scores across repeated runs, averaged over instances; lower variance indicates more consistent grading under sampling (Haldar & Hockenmaier, 2025a; Thakur et al., 2024). We also track end-to-end token usage.

For proprietary-model comparison, we evaluate GPT-5.1 through the OpenAI API (snapshot `gpt-5.1-2025-11-13`). Due to cost constraints, we run three trials on a representative subset: No-CoT, Full-CoT, Comparative, Quote-Forcing, Multi-Judge, G-Eval, Chain-of-Verification, and Overseer-guided Retry.

3.4. Adaptive routing with Mixture of Graders

The fixed-strategy sweep yields a judge-specific strategy–instance score matrix with entries $S_{q,s} = |\mathcal{R}_{q,s}|^{-1} \sum_r \hat{y}_{q,s,r}$, the mean extracted score across valid repeated runs; we also retain repeated-run variance $\sigma^2_{q,s}$ and mean token cost $C_{q,s}$. As a diagnostic upper bound, we define an in-hindsight oracle that selects the strategy maximizing $U_q(s) = -((S_{q,s} - y_q)/M_q)^2 - \lambda\sigma^2_{q,s}/M_q^2 - \gamma C_{q,s}/\bar{C}$, where y_q is the human reference label, M_q the benchmark maximum, and $\bar{C}$ normalizes token cost. The oracle is not deployable (it uses y_q), but it measures the headroom available to adaptive choice. MoG is trained from the full strategy-score matrix rather than by imitating hard oracle labels.

Router formulation. Mixture of Graders (MoG) learns a stochastic routing policy over grader strategies. For each instance q, the router receives only the problem statement and reference solution, $x_q = (\text{problem}_q, \text{reference}_q)$, and outputs a categorical distribution $p_\theta(\cdot \mid x_q) \in \Delta^{|\mathcal{S}|-1}$ over the 13 strategies. At evaluation and deployment time, MoG samples one strategy $\hat{s}_q \sim \text{Categorical}(p_\theta(\cdot \mid x_q))$ and uses its score $\hat{y}_q^{\text{MoG}} = S_{q,\hat{s}_q}$, so MoG is a sampled hard router rather than a deterministic weighted ensemble. The candidate solution and student-model identity are deliberately omitted from the router input, making MoG a problem/reference-conditioned policy rather than a full response-conditioned selector. This conservative design tests whether strategy preferences can be learned from problem structure alone and reduces the risk of learning artifacts tied to a particular student model's output style.

Training objective. Although MoG is evaluated by sampling a single strategy, we train it with a differentiable expected-score surrogate. The router minimizes a mixture-MSE loss $\mathcal{L}_{\text{MSE}}(\theta) = N^{-1}\sum_q(\tilde{y}_q - y_q)^2$, where $\tilde{y}_q = \sum_s p_\theta(s \mid x_q) S_{q,s}$ is the policy's expected score. Rather than imitating one hard oracle label when several strategies are nearly tied, the router increases probability mass on strategies whose scores move $\tilde{y}_q$ closer to the human label. Scores are normalized to $[0, 1]$ during training by dividing by the benchmark maximum and converted back to the native scale for reporting.

Router architecture and optimization. The router is a MODERNBERT-LARGE encoder with mean-pooled token embeddings and a 13-way linear head, with inputs truncated to 1024 tokens. We train with AdamW (learning rate 2×10^{-5} for the encoder, 10^{-3} for the head), batch size 16, two epochs of head-only warmup, and six epochs of joint fine-tuning. We train a separate router for each judge model because the strategy-score matrix, oracle headroom, and best fixed strategy are judge-specific.

Evaluation protocol. For Putnam-AXIOM-Grading, we evaluate MoG separately on candidate solutions from QWEN2.5-72B-MATH-INSTRUCT and MISTRAL-LARGE-INSTRUCT-2411. For each judge model and student-model response set, we train a separate router using 5-fold cross-validation over the 500 problem–candidate-solution instances (400 train / 100 held-out per fold), so every instance is evaluated once. For each fold, we compare MoG against the best fixed strategy selected by lowest MSE on the same held-out fold; this gives the fixed baseline access to the evaluation fold and is therefore conservative for MoG. Statistical significance is assessed with a paired bootstrap (10,000 resamples) over per-instance squared-error differences $d_q = (S_{q,\hat{s}_q} - y_q)^2 - (S_{q,c} - y_q)^2$ for comparator c and sampled route $\hat{s}_q$; negative values indicate that sampled MoG routing has lower MSE. To evaluate cross-generator transfer, we also train routers on one student-model response set and evaluate them on the other, using matched problem folds and the same response-blind router input.

4. Results and Analysis

4.1. Fixed strategies are useful but not sufficient

The fixed-strategy sweep shows that grader protocol materially affects both MSE and stability. Excluding MoG, the MSE-optimal fixed strategy varies across judge models and benchmarks (Tables 1 and 2). Comparative prompting is the strongest practical fixed default: it is single-call, consistently near the top, and achieves the best fixed-strategy MSE for DEEPSEEK-R1-DISTILL-QWEN-32B and GPT-5.1 on Putnam-AXIOM-Grading, as well as QWEN2.5-32B-INSTRUCT and LLAMA-3.1-70B-INSTRUCT on IMO-GradingBench. But it is not globally optimal. On Putnam, No-CoT is best for both Qwen judges and bullet-point scoring is best for LLAMA-3.1-70B-INSTRUCT; on IMO, Debate is best for QWEN2.5-72B-INSTRUCT and DEEPSEEK-R1-DISTILL-QWEN-32B, while Quote-Forcing is best among evaluated GPT-5.1 strategies.

Adding reasoning or pipeline complexity does not solve this variation. Multi-call strategies sometimes win on MSE, but they do not consistently dominate simpler single-call graders and often cost substantially more: G-Eval, CoVe, Debate, Multi-Judge, and Overseer-guided Retry use roughly 2–6× more tokens depending on benchmark and judge model (Tables 6 and 7). Their gains are also uneven: Debate has the lowest MSE for two open-weight judges on IMO but is high-variance, while G-Eval can be stable in some settings but often has high MSE. Conversely, low-variance strategies such as No-CoT or Multi-Judge often fail to minimize MSE. Fixed protocols therefore expose a three-way tradeoff between accuracy, stability, and cost: they are useful baselines, but no single fixed strategy resolves this tradeoff across judge models and benchmark distributions.

Table 1. MSE and variance ($\overline{\sigma^2}$) on Putnam-AXIOM-Grading, averaged across the Qwen2.5-72B and Mistral-Large-2411 student models. Both metrics are lower-is-better. Marking is based on each metric *within each model*: best is **bold**, second-best is underlined, third-best is marked with †. Some strategies in GPT-5.1 are omitted due to cost constraints.

		Qwen2.5-32B		**Qwen2.5-72B**		**DeepSeek-R1-32B**		**Llama-3.1-70B**		**GPT5.1**	
Type	**Strategy**	MSE	$\overline{\sigma^2}$	MSE	$\overline{\sigma^2}$	MSE	$\overline{\sigma^2}$	MSE	$\overline{\sigma^2}$	MSE	$\overline{\sigma^2}$
Ours	MoG	**.630**	.417	**.586**	.333	**.790**	.445	**.737**	.305	–	–
Baseline	No-CoT	$\underline{1.658}$	**.165**	$\underline{1.523}$	$\underline{.128}$	2.256	.353	2.054	$\underline{.213}$	.228	.019†
Single-Stage	Brief-CoT	2.024	.352	2.109	.233	2.045	.346	2.233	.350	–	–
	Full-CoT	2.368	.291	1.733	.140†	2.188	.318	2.223	.220†	.166	.021
	Structured-CoT	2.307	.443	2.935	.232	1.916	.318	2.391	.318	–	–
	Bullet-Point	1.788†	.235†	1.742	.161	$\underline{1.896}$	$\underline{.291}$	$\underline{1.753}$	.272	–	–
	Comparative	1.684†	.252	1.576†	.171	$\underline{1.792}$	.314	1.936†	.260	**.122**	.043
	Quote-Forcing	2.391	.485	1.924	.248	2.070	.353	2.099	.356	.203	.049
	Self-Critique	1.991	.343	2.075	.270	2.194	.310	2.162	.383	–	–
Multi-Step	G-Eval	3.349	.918	3.346	.730	2.323	.617	3.292	.577	.470	.122
	CoVe	2.147	.319	1.649	.180	2.111	.350	2.112	.287	.199	.034
	Multi-Judge	2.060	$\underline{.198}$	1.615†	**.091**	1.967	**.242**	2.163	**.160**	$\underline{.155}$	**.015**
	Debate	2.731	1.106	2.311	.606	2.440	.774	2.692	.811	–	–
	Overseer	2.438	.293	1.779	.140†	2.250	.302†	2.371	.227	$\underline{.155}$	$\underline{.019}$

4.2. Mixture of Graders improves over fixed strategy selection

The fixed-strategy sweep leaves substantial adaptive headroom. Under the primary MSE-aligned oracle setting ($\lambda = 1, \gamma = 0$), the oracle-selected strategy differs from the validation-selected best fixed strategy in 87.8% of held-out Putnam model-instance cases and 70.1% of IMO cases, or 75.9% overall (Table 3). This translates into large oracle gains: oracle selection reduces MSE by 68.0%–80.2% on Putnam-AXIOM-Grading and 11.3%–29.6% on IMO-GradingBench relative to the best fixed strategy (Table 4). Sensitivity sweeps over the stability penalty λ and token-cost penalty γ show that this headroom persists across objective choices, and oracle selections do not collapse to a single fixed protocol (Appendix I).

MoG converts this headroom into deployable gains. On Putnam-AXIOM-Grading, averaged across the QWEN2.5-72B-MATH-INSTRUCT and MISTRAL-LARGE-INSTRUCT-2411 student response sets, MoG achieves the lowest MSE for every open-weight judge model (Table 1). Relative to the strongest fixed strategy, MoG reduces MSE by more than half for QWEN2.5-32B-INSTRUCT, QWEN2.5-72B-INSTRUCT, DEEPSEEK-R1-DISTILL-QWEN-32B, and LLAMA-3.1-70B-INSTRUCT, recovering roughly two-thirds to three-quarters of the available Putnam oracle headroom. In the pooled cross-judge view (Figure 1), MoG beats the best fixed strategy for every judge, with all paired bootstrap confidence intervals excluding zero by at least 3.6σ and the largest relative MSE reductions on the Qwen judges. Because the best fixed baseline is selected by lowest MSE on the same held-out fold, these gains are measured against a conservative comparator.

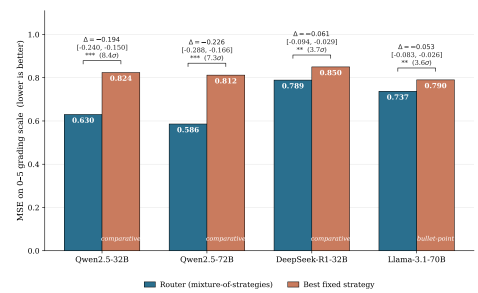

Figure 1. **MoG beats the best fixed prompting strategy for every open-weight judge.** Results use pooled 5-fold CV on $Q_{\text{eff}} = 1000$ held-out (problem, student-model) instances per judge from the original Putnam-AXIOM-Grading responses and a second Mistral-generated response set. Bars show MSE on the 0–5 grading scale; lower is better. The best fixed strategy is the per-judge pooled-validation winner (*comparative* for Qwen/DeepSeek, *bullet-point* for LLAMA-3.1-70B). Brackets report $\Delta = \mathrm{MSE}_{\text{router}} - \mathrm{MSE}_{\text{fixed}}$, 95% percentile bootstrap CI over 10,000 resamples, and effect size in bootstrap SEs. All CIs exclude zero by $\geq 3.6\sigma$; gains are largest for the Qwen judges. $^{**}p < 10^{-3}$; $^{***}p < 10^{-4}$.

4.3. Routing generalizes across student-model outputs

A natural concern with response-blind routing is that the learned policy may depend on idiosyncrasies of the student-model response distribution. To test this, we train MoG on Putnam-AXIOM-Grading instances generated by a QWEN2.5-72B-MATH-INSTRUCT student model and evaluate it without retraining on the parallel MISTRAL-LARGE-INSTRUCT-2411 response set, using the same 500 problems and the same QWEN2.5-32B-INSTRUCT judge. Because the router never sees the candidate solution, this directly tests whether problem/reference-conditioned strategy preferences transfer across student-output distributions.

Under the expected-router deployment mode,

$$\tilde{y}_q = \sum_s p_\theta(s \mid x_q) S_{q,s},$$

the transferred router achieves MSE 0.538 on Mistral responses, outperforming the uniform mixture over strategies (0.623) and all fixed strategies in point estimate, including Comparative prompting (0.589), the strongest fixed default (Table 5). The paired gap against Comparative is $\Delta\text{MSE} = -0.050$ with 95% bootstrap CI $[-0.113, +0.013]$ and one-sided $p = 0.059$; the router beats Comparative in 94.1% of paired bootstrap resamples. It also significantly outperforms 12 of the 13 fixed strategies and the uniform mixture ($p < 0.05$), and recovers a median 11.5% of the oracle headroom.

4.4. Analysis of Baseline Strategies

The single-call baselines form a surprisingly strong reference point. No-CoT is the most stable strategy on every model except Llama-3.1-70B for IMO, achieving the lowest variance on Qwen2.5-32B ($\sigma^2 = .169$ IMO, .165 Putnam) and GPT-5.1 ($\sigma^2 = .047$ IMO), and remaining within striking distance of the best MSE on Putnam—at a fraction of the token budget of any pipeline. Forgoing rationales removes the largest source of stochasticity (free-form CoT traces), which directly suppresses run-to-run variance and prevents the judge from talking itself into different scores on repeated samples. Among the single-turn CoT variants, Comparative is the clear winner on agreement, attaining the best or second-best MSE on 8 of 10 (model, benchmark) cells—including the top score for Qwen2.5-32B and Llama-3.1-70B on IMO and the top GPT-5.1 result on Putnam (.122). Bullet-Point is the consistent runner-up, plausibly because its terse criterion-wise format constrains the reasoning surface and so retains some of No-CoT's stability while allowing limited rubric-aligned deliberation. The pattern across the remaining single-turn variants (Brief-, Full-, Structured-CoT, Quote-Forcing, Self-Critique) is that adding unconstrained or rigidly templated reasoning generally raises σ^2 without a commensurate MSE gain, indicating that the *structure* of elicited reasoning matters more than its presence: strategies that keep the judge focused on holistic, reference-anchored assessment (Comparative, Bullet-Point) outperform those that either fragment the task or let rationales drift.

5. Limitations

We evaluate four open-weight judges and one proprietary judge (GPT-5.1), and train MoG only on the open-weight judges; rankings and routing gains may differ for other architectures, especially reasoning-specialized models beyond DeepSeek-R1. Each of the 13 grader protocols is instantiated with a single prompt template, so our results characterize the strategies as implemented rather than the full space of possible instantiations.

MoG itself has three further limitations. First, training the router requires running every candidate grader on every training instance to build the strategy–instance score matrix; inference cost matches a single grader, but the supervision signal is expensive to obtain. Second, the router is response-blind and judge-specific by design—it conditions only on the problem and reference solution, and we train one router per judge—so it cannot react to features of the specific candidate solution and we do not explore a shared cross-judge router. Third, our adaptive-routing experiments focus on Putnam-AXIOM-Grading; whether the approach helps without reference solutions or outside competition mathematics is left to future work.

6. Conclusion

We study LLM-as-a-judge for difficult mathematical grading and ask whether a single fixed grader protocol is the right abstraction. Across two competition-mathematics benchmarks and five judge models, no fixed protocol dominates: comparative prompting is the strongest fixed default on average, but an in-hindsight oracle over 13 strategies selects a different protocol on roughly 76% of instances and obtains substantially lower MSE. Pipeline complexity does not resolve this—G-Eval, Debate, CoVe, and multi-judge ensembling are not consistently better than well-chosen single-call prompts despite costing several times more tokens—and unconstrained chain-of-thought tends to inflate variance without improving agreement.

Motivated by this gap, we introduce **Mixture of Graders**, a MODERNBERT-LARGE router trained on the fixed-strategy score matrix and human labels but conditioning only on the problem and reference solution at inference. On Putnam-AXIOM-Grading, MoG attains the lowest MSE on every open-weight judge we test, with relative reductions up to 27.8% over the best fixed strategy, and a router trained on Qwen student responses outperforms every fixed strategy when applied to Mistral responses without retraining. The right abstraction for difficult mathematical grading is not a universal prompt but a learned mixture: pair strong fixed defaults with a router that decides, per problem, which grader to call.

Impact Statement

This work advances the reliability of LLM-based evaluation for mathematical reasoning, with implications for how frontier model capabilities are measured and compared. Benchmark scores increasingly drive decisions about which models are deployed and which training methods are pursued; if those scores depend on arbitrary grading protocol choices, capability comparisons may be systematically distorted. By demonstrating that no fixed grading protocol dominates and that adaptive strategy selection reduces grading error, this work encourages the field to treat evaluation methodology as a first-class design choice rather than a fixed background assumption.

The Mixture of Graders framework could reduce reliance on expensive human grading for mathematical assessment, with benefits for scalable evaluation pipelines. However, automated graders—even improved ones—may introduce systematic biases that differ from human judgment in ways that are difficult to detect, particularly on novel problem types outside the training distribution. Practitioners deploying such systems for high-stakes assessment (e.g., educational evaluation or alignment feedback) should treat automated grades as estimates with quantified uncertainty rather than ground truth. The benchmarks and code released with this work are intended to support further scrutiny of grading reliability.

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A. IMO Table

Table 2. Agreement and stability of judge strategies on **IMO-GradingBench**. We report Mean Squared Error against human grades (MSE, lower is better) and mean within-condition variance (σ^2, lower is better). Best results per model are **bolded**; second-best are underlined. Strategies are grouped by type: single-turn baselines, single-turn CoT variants, and multi-step pipelines. Some strategies in GPT-5.1 are omitted due to cost constraints.

		Qwen2.5-32B		**Qwen2.5-72B**		**DeepSeek-R1-32B**		**Llama-3.1-70B**		**GPT-5.1**	
Type	**Strategy**	MSE	σ^2	MSE	σ^2	MSE	σ^2	MSE	σ^2	MSE	σ^2
Baseline	No-CoT	14.981	**.169**	18.569	.167	16.998	.280	18.417	**.392**	8.490	**.047**
Single-Turn	Brief-CoT	13.781	.414	19.152	.199	15.573	.313	16.006	.522	–	–
	Full-CoT	15.154	.332	19.831	.192	16.870	.282	14.861	.491	7.780	.225
	Structured-CoT	15.173	.420	20.063	.185	15.951	.319	15.462	.567	–	–
	Bullet-Point	12.752	.339	17.413	.285	14.429	.289	12.099	.776	–	–
	Comparative	**11.457**	.327	14.739	.216	15.238	.313	**9.443**	.866	7.427	.251
	Quote-Forcing	15.110	.346	19.357	.194	16.150	.291	14.884	.570	**7.269**	.119
	Self-Critique	14.090	.382	19.233	.176	16.255	.272	15.643	.618	–	–
Multi-Step	G-Eval	14.917	.381	19.967	**.059**	15.674	.370	18.617	.428	8.024	.231
	CoVe	14.271	.397	19.183	.205	16.561	.313	14.163	.558	8.115	.319
	Multi-Judge	14.393	.248	19.315	.124	16.620	**.230**	14.226	.401	7.745	.250
	Debate	13.749	.388	**11.785**	.534	**13.898**	.700	15.404	1.335	–	–
	Overseer	15.154	.332	19.849	.189	17.113	.275	15.154	.509	9.385	.220

B. Adaptive Strategy Headroom Results

Table 3. How often the validation-selected best fixed strategy matches or differs from the oracle-selected strategy under the primary MSE-aligned setting $\lambda = 1, \gamma = 0$.

Benchmark	Model	Best fixed	Match	Different
Putnam-AXIOM-Grading	DeepSeek-R1-32B	Comparative	11.3%	88.7%
	Llama-3.1-70B	Bullet-point	15.6%	84.4%
	Qwen2.5-32B	Comparative	9.1%	90.9%
	Qwen2.5-72B	Comparative	12.6%	87.4%
IMO-GradingBench	DeepSeek-R1-32B	Debate	14.5%	85.5%
	Llama-3.1-70B	Comparative	37.6%	62.4%
	Qwen2.5-32B	Comparative	16.5%	83.5%
	Qwen2.5-72B	Debate	51.1%	48.9%

Table 4. MSE-aligned strategy-selection headroom on held-out instances. ΔU reports the utility improvement of the oracle selector over the validation-selected best fixed strategy under the primary setting $\lambda = 1, \gamma = 0$. MSE is reported on the native benchmark scale: 0–5 for Putnam and 0–7 for IMO. Higher ΔU and Pearson are better; lower MSE is better.

Benchmark	Model	Best fixed	ΔU	Oracle MSE	Fixed MSE	MSE red.	Oracle ρ	Fixed ρ
Putnam	DeepSeek-R1-32B	Comparative	.041	.392	1.225	68.0%	.924	.750
	Llama-3.1-70B	Bullet-point	.033	.264	.971	72.8%	.933	.726
	Qwen2.5-32B	Comparative	.039	.288	1.197	75.9%	.932	.731
	Qwen2.5-72B	Comparative	.037	.221	1.115	80.2%	.945	.751
IMO	DeepSeek-R1-32B	Debate	.051	11.352	13.672	17.0%	.680	.521
	Llama-3.1-70B	Comparative	.041	6.633	8.367	20.7%	.806	.570
	Qwen2.5-32B	Comparative	.067	7.878	11.185	29.6%	.763	.496
	Qwen2.5-72B	Debate	.029	9.968	11.233	11.3%	.708	.457

C. MoG Transfer Ablation

Table 5. Router transfer to a held-out student model on Putnam-AXIOM-Grading. A router trained on Qwen-2.5-32B student responses is evaluated, without retraining, on Mistral student responses graded by the same Qwen-2.5-32B judge. We report MSE of the routed score against human grades on the native 0–5 Putnam total-score scale, with 95% paired bootstrap confidence intervals over $n = 500$ problems ($B = 10{,}000$ replicates). The bottom block reports both router deployment modes alongside the uniform mixture and the oracle upper bound, plus the paired contrast ΔMSE against the strongest fixed default, Comparative (p is one-sided). † marks rows where Router (expected) significantly outperforms the strategy on MSE (paired bootstrap, $p < 0.05$). The best non-oracle MSE is **bolded**.

Type	Strategy	MSE ↓	95% CI	ΔMSE vs. Comp. (p)
Baseline	No-CoT†	0.7004	[0.6155, 0.7900]	—
Single-Turn	Brief-CoT†	0.8266	[0.7282, 0.9351]	—
	Full-CoT†	1.2334	[1.1099, 1.3568]	—
	Structured-CoT†	0.9810	[0.8467, 1.1207]	—
	Bullet-Point†	0.6845	[0.5928, 0.7872]	—
	Comparative	*0.5886*	[0.5099, 0.6696]	—
	Quote-Forcing†	0.9659	[0.8430, 1.0974]	—
	Self-Critique†	0.7301	[0.6336, 0.8324]	—
Multi-Step	G-Eval†	1.4807	[1.3115, 1.6635]	—
	CoVe†	1.0202	[0.9180, 1.1248]	—
	Multi-Judge†	0.9583	[0.8543, 1.0673]	—
	Debate†	0.9436	[0.8231, 1.0710]	—
	Overseer†	1.2566	[1.1349, 1.3865]	—
Routed	Uniform mixture	0.6228	[0.5444, 0.7066]	—
	Router (expected)	**0.5383**	[0.4713, 0.6105]	-0.0502 (p=0.059)
Upper bound	Oracle	0.1505	[0.1213, 0.1830]	—

Headroom captured by Router (expected): median 11.5% of the gap between the best fixed strategy and the oracle (95% CI $[-3.2\%,\ +23.7\%]$); $P(\text{Router} < \text{best fixed}) = 0.941$ across paired bootstrap replicates.

D. Token Calculation Formulas

Token Cost Summary

Per call:

$$T_{\text{call}} = T_{\text{in}}(\text{call}) + T_{\text{out}}(\text{call}) \tag{1}$$

Per strategy (summed over problems i):

$$T_{\text{strategy}} = \sum_i T_{\text{strategy}}(i) \tag{2}$$

Per-problem token costs by strategy:

Single-turn (`no_cot`, `brief_cot`, `structured_cot`, `full_cot`, `self_critique`, `bullet_point`, `comparative`, `quote_forcing`):

$$T = T_{\text{in}}(\text{prompt}) + T_{\text{out}}(\text{output}) \tag{3}$$

G-Eval (two-stage checklist then scoring):

$$T = \big[T_{\text{in}}(\text{step}_1) + T_{\text{out}}(\text{checklist})\big] + \big[T_{\text{in}}(\text{step}_2(\text{checklist})) + T_{\text{out}}(\text{output})\big] \tag{4}$$

Debate (pro, con, then arbiter):

$$\begin{aligned} T = &\big[T_{\text{in}}(\text{pro}) + T_{\text{out}}(\text{pro\_arg})\big] \\ &+ \big[T_{\text{in}}(\text{con}) + T_{\text{out}}(\text{con\_arg})\big] \\ &+ \big[T_{\text{in}}(\text{arbiter}(\text{pro\_arg, con\_arg})) + T_{\text{out}}(\text{output})\big] \end{aligned} \tag{5}$$

CoVe (draft, verify, answer, finalise):

$$\begin{aligned} T = &\big[T_{\text{in}}(\text{draft}) + T_{\text{out}}(\text{draft})\big] \\ &+ \big[T_{\text{in}}(\text{verify}(\text{draft})) + T_{\text{out}}(\text{questions})\big] \\ &+ \big[T_{\text{in}}(\text{answer}(\text{questions})) + T_{\text{out}}(\text{answers})\big] \\ &+ \big[T_{\text{in}}(\text{final}(\text{draft, Q\&A})) + T_{\text{out}}(\text{output})\big] \end{aligned} \tag{6}$$

Multi-Judge (three persona judges then chair):

$$T = \sum_{j=1}^{3}\big[T_{\text{in}}(\text{persona}_j) + T_{\text{out}}(\text{judgment}_j)\big] + T_{\text{in}}(\text{chair}(j_1, j_2, j_3)) + T_{\text{out}}(\text{output}) \tag{7}$$

Overseer-Retry (summed over all iterations k):

$$T = \sum_k \Big(\big[T_{\text{in}}(\text{judge}_k) + T_{\text{out}}(\text{judge\_out}_k)\big] + \big[T_{\text{in}}(\text{overseer}_k) + T_{\text{out}}(\text{overseer\_out}_k)\big]\Big) \tag{8}$$

where k iterates over all entries in `overseer_checks`.

E. Token Usage Analysis

Table 6. Total token usage of judge strategies across models on **Putnam-AXIOM-Grading**. We report total tokens consumed per strategy per model. Highest tokens per model are **bolded**; second-highest are <u>underlined</u>. Strategies are grouped by type: baseline, single-turn variants, and multi-step pipelines.

Type	Strategy	DeepSeek-R1-32B Tokens	Qwen2.5-32B Tokens	Qwen2.5-72B Tokens	Llama-3.1-70B Tokens
Baseline	No-CoT	866,526	850,780	851,797	844,746
Single-Turn	Brief-CoT	856,775	855,903	855,932	849,895
	Full-CoT	1,037,450	1,006,790	1,009,740	1,001,598
	Structured-CoT	988,835	933,031	932,164	928,397
	Bullet-Point	886,734	890,603	890,824	885,537
	Comparative	958,600	895,427	894,719	888,687
	Quote-Forcing	983,522	891,478	896,288	887,367
	Self-Critique	1,103,194	916,705	918,074	914,197
Multi-Step	G-Eval	2,363,471	1,960,149	1,905,160	2,085,316
	CoVe	5,125,495	3,774,706	3,799,769	3,860,194
	Multi-Judge	**6,070,101**	**4,804,855**	**4,937,384**	<u>4,825,306</u>
	Debate	4,744,275	3,532,677	4,190,834	3,695,543
	Overseer	<u>4,545,332</u>	<u>2,884,088</u>	<u>3,499,382</u>	**5,407,469**

Table 7. Total token usage of judge strategies across models on **IMO-GradingBench**. We report total tokens consumed per strategy per model. Highest tokens per model are **bolded**; second-highest are <u>underlined</u>. Strategies are grouped by type: baseline, single-turn variants, and multi-step pipelines.

Type	Strategy	DeepSeek-R1-32B Tokens	Qwen2.5-32B Tokens	Qwen2.5-72B Tokens	Llama-3.1-70B Tokens
Baseline	No-CoT	5,132,342	4,440,810	4,440,810	4,424,912
Single-Turn	Brief-CoT	4,937,699	4,497,078	4,491,580	4,484,273
	Full-CoT	5,557,631	5,013,052	5,132,245	5,012,472
	Structured-CoT	5,418,253	4,886,272	4,833,874	4,805,231
	Bullet-Point	5,350,817	4,760,237	4,710,583	4,693,688
	Comparative	5,298,131	4,796,308	4,735,847	4,707,437
	Quote-Forcing	5,346,101	4,878,911	4,854,241	4,828,665
	Self-Critique	5,394,857	4,791,587	4,791,586	4,797,907
Multi-Step	G-Eval	11,221,644	9,251,876	9,158,026	9,152,668
	CoVe	20,402,280	14,964,140	15,242,634	14,813,446
	Multi-Judge	**24,068,464**	**20,357,440**	**21,004,026**	**20,412,120**
	Debate	17,228,546	14,088,592	14,980,337	14,175,135
	<u>Overseer</u>	<u>16,709,961</u>	<u>9,853,854</u>	<u>10,179,871</u>	<u>10,510,512</u>

F. Comparative Prompting Ablations

Given comparative prompting's consistently dominant performance we conduct targeted ablations to identify its active ingredients. We test five variants against the original (A0) on all four open-source models across both benchmarks (Table 8).

Template ablations We test three modifications to the comparative template: (A1) **Comparative no-compare** uses the same four-step template but replaces "compare to ground truth" with "evaluate the solution's correctness," while keeping the ground truth in context. This tests whether explicitly directing attention to the reference is the active ingredient. (A2) **Difference-only** strips the template to a minimal prompt: "List the differences between the model output and the ground truth, then score." This tests whether the bare act of difference-finding suffices without the template structure. (A4) **Comparative shuffled** uses the same four components as A0 but in a randomized order (error analysis → scoring rationale → approach comparison → differences). This tests whether the specific progressive-narrowing ordering matters.

Attention-direction ablations We test two variants that redirect the judge's focus: (A5) **Quote-forcing reference** requires the judge to cite direct quotes from the *ground truth* rather than the student solution, testing whether anchoring attention specifically to the reference drives performance. (A3) **Reference checklist** auto-generates 3–5 binary Yes/No verification questions from the ground truth, then has the judge answer them before scoring—similar to G-Eval but with checklist items derived from the reference rather than freely generated.

Results The ablation results (Table 8) fail to support any single-ingredient explanation for comparative prompting's effectiveness:

- **Attention anchoring is not the driver.** If explicitly directing attention to the reference were the key mechanism, A5 (quote-forcing reference) should outperform standard quote-forcing and approach A0. Instead, A5 consistently underperforms A0 across both benchmarks. Conversely, A1 performs competitively despite *not* explicitly referencing the ground truth in its instructions, suggesting that the reference need only be *present in context*, not explicitly called out.
- **Template structure is helpful but not sufficient.** A2 (difference-only), which removes all template structure, performs comparably to A0 on Putnam-AXIOM-Grading (matching or exceeding it for Qwen-72B) but degrades on IMO-GradingBench, indicating that structure contributes but is not the sole factor.
- **Task framing ("compare" vs. "evaluate") does not explain the gap.** A1 and A2 differ precisely on this dimension—A1 frames the task as evaluation while A2 frames it as difference-finding—yet perform similarly, ruling out cognitive framing as the primary mechanism.
- **Component ordering has limited effect.** A4 (shuffled order) performs as well as or better than A0 on Putnam-AXIOM-Grading, suggesting the progressive-narrowing structure of the original template is not critical. On IMO-GradingBench, A4 shows more variable results but remains competitive.
- **Structured reference decomposition hurts.** A3 (reference checklist) consistently underperforms across both benchmarks, mirroring G-Eval's poor showing in the main experiments—suggesting that decomposing the reference into binary checks destroys holistic reasoning.

Interpretation The pattern that emerges is that *all high-performing variants share a common property*: they allow the judge to holistically reason about both solutions with minimal output constraints. The strategies that degrade performance—A3 (reference checklist) and A5 (quote-forcing reference)—are precisely those that impose rigid intermediate output requirements, forcing the judge to fragment its reasoning into constrained artifacts (binary answers, quoted evidence) before scoring. We term this active ingredient *evaluation-focused holistic reasoning*: the prompt keeps the judge's attention on the substantive task of assessment rather than diverting it to mechanical subtasks. This also explains why G-Eval and Debate perform poorly in the main experiments—both impose intermediate reasoning structures that fragment the evaluation process.

Table 8. Ablation results for comparative prompting on Putnam-AXIOM-Grading and IMO-GradingBench. We report Pearson correlation with human grades (ρ, higher is better) and mean within-condition variance (σ^2, lower is better) across five independent trials. Best results per model–metric are **bolded**; second-best are underlined. Strategies are ordered by conceptual grouping: original comparative, template ablations, and attention-direction ablations.

	Pearson (ρ) $\uparrow$				Variance (σ^2) $\downarrow$			
Strategy	Qwen-32B	Qwen-72B	DS-R1-32B	Llama-70B	Qwen-32B	Qwen-72B	DS-R1-32B	Llama-70B
(a) Putnam-AXIOM-Grading								
BASELINE								
Comparative (A0)	.715	.733	.725	.677	.239	.179	.316	.293
TEMPLATE ABLATIONS								
Comp. no compare (A1)	.716	.714	.728	.681	**.219**	.184	.293	**.263**
Difference only (A2)	.718	.736	.704	.688	.254	**.158**	.322	.261
Comp. shuffled (A4)	**.732**	**.761**	**.740**	**.689**	.251	.165	**.279**	.297
ATTENTION-DIRECTION ABLATIONS								
Quote-force ref. (A5)	.690	.718	.684	.654	.361	.185	.334	.368
Ref. checklist (A3)	.678	.614	.660	.661	.419	.390	.427	.408
(b) IMO-GradingBench								
BASELINE								
Comparative (A0)	.493	**.505**	**.490**	.392	.332	.368	.317	.555
TEMPLATE ABLATIONS								
Comp. no compare (A1)	**.498**	.427	.465	.451	.367	**.252**	.339	**.429**
Difference only (A2)	.478	.460	.436	.463	**.265**	.265	**.257**	.730
Comp. shuffled (A4)	.486	.475	.460	**.465**	.352	.445	.338	.750
ATTENTION-DIRECTION ABLATIONS								
Quote-force ref. (A5)	.452	.450	.442	.429	.455	.387	.326	1.04
Ref. checklist (A3)	.341	.241	.411	.399	.398	.278	.435	.686

G. Grading Rubrics Used

Reasoning-quality rubric (MODEL_OUTPUT vs. GROUND_TRUTH_SOLUTION).

Task: Evaluate the `MODEL_OUTPUT`'s *reasoning quality* against the `GROUND_TRUTH_SOLUTION`. Focus *only* on reasoning quality (not writing style).

Criteria (each scored 0–1; 0.5 allowed):

1. **Logical Coherence (0–1)**: Are steps locally valid and globally consistent?
2. **Faithfulness to Task (0–1)**: Does the reasoning address the question (no hallucinated subgoals or irrelevant paths)?
3. **Methodological Alignment (0–1)**: Is the approach compatible with (or reasonably equivalent to) the canonical method in the ground truth? If different, is it still valid?
4. **Intermediate Correctness (0–1)**: Are key intermediate results (derivations, lemmas, sub-calculations) correct?
5. **Error Awareness / Self-Correction (0–1)**: Does the reasoning notice and repair mistakes or check edge cases? (If model does not make any nontrivial mistakes, do not penalize here).

Scoring: Final score is the sum across the five criteria (0–5; half-points allowed).

Figure 2. Rubric used by judge LLM to evaluate student solution for Putnam problems.

IMO-style grading rubric (0–7).

Task: Grade the student's solution on the standard IMO 0–7 scale. Output a single integer score in $\{0, 1, \ldots, 7\}$.

Scale definitions:

7 Completely correct solution.
6 Correct solution with very minor slips (e.g., a sign error that doesn't affect the logic).
5 Main theorem/logic is established, but a significant case or detail is missed.
4 Significant progress toward the solution (e.g., reducing the problem to a known solved state), but main proof incomplete.
3 Non-trivial partial results that are steps toward the solution.
2 Some relevant formulas or observations, but no significant progress.
1 Fundamental misunderstanding or only trivial observations.
0 Completely incorrect or blank.

Output format: Return only the integer score (no extra text).

Figure 3. Rubric used by judge LLM to evaluate student solution for IMO problems.

H. Deepseek R1 Analysis

Section 4 established that DeepSeek-R1-Distill-Qwen-32B is a consistent exception to the pattern of CoT degrading judge reliability: on Putnam-AXIOM-Grading, structured-CoT improves agreement by 9.4 points over no-CoT, and the model benefits from CoT across both benchmarks regardless of structure. We attributed this to the model's reinforcement-learning training on reasoning traces, which leaves it "undertrained for the terse responses required by direct scoring." Here we present a fine-grained analysis of *how* the Full-CoT and No-CoT conditions differ for this model, based on a comparison of all 500 Putnam-AXIOM-Grading outputs from a single representative trial.

DeepSeek-R1 cannot suppress reasoning. The most fundamental observation is that DeepSeek-R1-Distill-Qwen-32B generates extensive chain-of-thought reasoning in *both* conditions. Even under the No-CoT prompt—which instructs the model to output only a JSON score with no reasoning—497 of 500 outputs (99.4%) contain a </think> token demarcating an internal reasoning scratchpad. These scratchpads average 2,016 characters, only 10% shorter than the Full-CoT scratchpads (2,243 characters). After the </think> token, the No-CoT outputs contain only the JSON score block ($\sim$230 characters). This confirms the behavioral observation in Section 4.1: for reasoning-optimized models, "direct scoring" is not just less effective but behaviorally unnatural—the model *cannot help but deliberate*.

The critical difference lies not in whether the model reasons, but in what happens *after* the scratchpad. Under Full-CoT, 431 of 500 outputs (86.2%) exhibit a **two-phase structure**: a free-form scratchpad (enclosed within the model's internal think tokens), followed by a structured <THINKING> summary that distills the scratchpad's observations into per-criterion assessments, followed by the JSON score. Under No-CoT, the model proceeds directly from its scratchpad to the JSON score, with no intermediate consolidation step.

The distillation step corrects leniency on surface-level criteria. This architectural difference has measurable consequences. Comparing per-component score means across the two conditions reveals an asymmetric pattern: No-CoT assigns higher scores on *methodological alignment* ($+0.10$ shift, from 0.55 to 0.65) and *logical coherence* ($+0.05$ shift), while assigning *lower* scores on *error awareness* (-0.12 shift). The first two criteria are precisely those assessable from a surface reading of the student solution: "Does the method seem reasonable?" and "Do the steps flow logically?" The No-CoT scratchpad tends to answer these affirmatively based on the student's internal consistency, without weighing the ground truth.

The error awareness shift is a striking counterpoint: No-CoT is actually *harsher* on this criterion (mean 0.16 vs. 0.28 under Full-CoT), but this harshness appears nearly random—the Pearson correlation between No-CoT's error awareness scores and human total scores is only $r = 0.39$, compared to $r = 0.64$ under Full-CoT, the largest per-component gap we observe ($\Delta r = +0.25$). Substantial gaps also appear for logical coherence ($\Delta r = +0.08$) and methodological alignment ($\Delta r = +0.07$).

No-CoT compresses scores toward the midrange. The net effect of these per-component shifts is a compression of the total score distribution. Under No-CoT, 65.3% of scores fall in the 1.5–4.0 range, compared to 48.7% under Full-CoT and 48.4% for human grades. Correspondingly, No-CoT produces fewer extreme scores: only 15.9% of scores are ≤ 1 (vs. 21.2% for Full-CoT, 36.9% for humans) and only 17.8% are ≥ 4.5 (vs. 29.2% for Full-CoT). The reduced spread (standard deviation 1.37 vs. 1.56 for Full-CoT, 1.43 for humans) directly reduces Pearson correlation: on the 415 instances where both conditions parse successfully, Full-CoT achieves $r = 0.69$ versus $r = 0.64$ for No-CoT.

Qualitative mechanism: noticing errors without penalizing them. The quantitative patterns are explained by a consistent qualitative mechanism visible in individual outputs (Figures 4–6). In the No-CoT condition, the model's scratchpad frequently *identifies* discrepancies with the ground truth but then *rationalizes them away* in the same breath, defaulting to a lenient assessment of the student solution's internal logic. Three illustrative failure modes emerge:

1. **Internal-consistency bias** (Figure 4). The scratchpad notes that the student derives $I_k = I_{5-k}$ instead of the correct I_{4-k}, but then writes: "However, the steps are locally valid and globally consistent *within its own reasoning*, so I'll give a 1." The judge evaluates the solution as a standalone argument rather than against the reference.

2. **Premise-blindness** (Figure 5). The student's internal mathematics is correct (a quadratic is properly solved), but the entire argument rests on a false premise contradicted by the ground truth. The No-CoT scratchpad never confronts this,

writing: "The calculations are accurate. So, Intermediate Correctness is 1." The Full-CoT scratchpad immediately cross-references: "doesn't align with the ground truth, which shows all u's are zero."

3. **Failure to propagate errors across criteria** (Figure 6). The No-CoT scratchpad notices that the student's answer is wrong but confines the penalty to a single criterion (Intermediate Correctness), leaving Logical Coherence, Faithfulness, and Methodological Alignment at full marks. The Full-CoT <THINKING> summary propagates the error backward, recognizing that an incorrect conclusion undermines the coherence and faithfulness of the reasoning that led to it.

In each case, the Full-CoT condition avoids the failure because its two-phase structure—scratchpad followed by <THINKING> summary—forces the model to *consolidate* its observations into rubric-calibrated judgments. The scratchpad serves as exploratory analysis; the summary serves as deliberate assessment. Without this distillation step, the model's initial impressions pass through to the final score unchecked.

Implications. These findings refine the claim in Section 4.1 that reasoning-trained models benefit from CoT because they are "undertrained for terse responses." The mechanism is more specific: DeepSeek-R1 deliberates regardless of the prompt, but the *quality of the transition from deliberation to scoring* depends on whether the prompt elicits an explicit output-side reasoning structure. The Full-CoT prompt does not make the model think more—the scratchpads are similar in length and content—but it provides a scaffold for converting messy, exploratory reasoning into calibrated judgments. For practitioners using reasoning-specialized models as judges, this suggests that the key design choice is not whether to allow CoT (the model will reason anyway) but whether to require a structured output format that forces distillation of that reasoning before scoring.

Problem (Putnam #227)

Let $p(x) = 2 + 4x + 3x^2 + 5x^3 + 3x^4 + 4x^5 + 2x^6$. For k with $0 < k < 5$, define $I_k = \int_0^\infty \frac{x^k}{p(x)}\,dx$. For which k is I_k smallest?

Ground Truth Solution (abridged)

Substitute $x = 1/t$ to show $I_k = I_{4-k}$. Then $I_k = (I_k + I_{4-k})/2 = \int_0^\infty \frac{(x^k+x^{4-k})/2}{p(x)}\,dx > \int_0^\infty \frac{x^2}{p(x)}\,dx = I_2$ by AM–GM, since $(x^k + x^{4-k})/2 \geq x^2$. Answer: $k = \boxed{2}$.

Student Solution (abridged)

Notes $p(x) = x^6 p(1/x)$ (correct). Claims $I_k = I_{5-k}$ (should be I_{4-k}). Approximates $p(x) \approx 23$ near $x = 1$, argues minimum at $k = 5/2$. Assumes k must be integer (not stated in problem). Checks $k = 2$ vs. $k = 3$, concludes $k = 2$. Gets the right answer through flawed reasoning.

No-CoT Judge (score: 4.5 / 5)

"The model's reasoning seems to flow well [. . .] I notice that the model assumes symmetry around 5, not 4 as in the ground truth. However, the steps are locally valid and globally consistent within its own reasoning, so I'll give a 1."

"The model correctly identifies $p(x)$ as symmetric, but the substitution in the ground truth relates I_k to I_{4-k}, not I_{5-k}. The model's symmetry is slightly off. However, the main conclusion k=2 is correct, so intermediate steps are mostly correct. Score: 1."

"Both methods are valid but different. The model's approach is reasonable, so alignment is present. Score: 1."

LC: 1 — FT: 1 — MA: 1 — IC: 1 — EA: 0.5 — **Total: 4.5**

Full-CoT Judge (score: 1.0 / 5)

Scratchpad:
"Wait, that's a discrepancy. The ground truth uses $4-k$, but the model says $5-k$. Hmm, that might be an error."

"Wait, but why does the model assume k must be an integer? The problem states $0 < k < 5$, but k doesn't have to be an integer [. . .] That could be a mistake."

⟨THINKING⟩ summary:
"Methodological Alignment (0): The approach differs from the ground truth [. . .] leading to invalid steps."
"Intermediate Correctness (0): Key steps, like the substitution and integer assumption, are incorrect, undermining the reasoning."

LC: 0.5 — FT: 0.5 — MA: 0 — IC: 0 — EA: 0 — **Total: 1.0**

Figure 4. **"Locally valid within its own reasoning" (Human: 1.0).** Both judges notice that the student derives $I_k = I_{5-k}$ instead of the correct I_{4-k}. The No-CoT judge excuses this: because the steps are internally consistent, it awards full marks on four criteria. The Full-CoT judge's scratchpad flags the same discrepancy ("*Wait, that's a discrepancy*") and the unjustified integer assumption, then the structured <THINKING> summary consolidates these into zero-score penalties on methodology and correctness—matching the human grade exactly. The key failure mode: **without a distillation step, No-CoT evaluates the student solution as a standalone argument rather than against the reference standard.**

Problem (Putnam #92)

Suppose $u_0, u_1, u_2, \ldots$ is a sequence of real numbers with $u_n = \sum_{k=1}^{\infty} u_{n+k}^2$ for all $n \geq 0$. If $\sum u_n$ converges, find the maximum value of u_k for all k.

Ground Truth (abridged)

Shows $\{u_n\}$ is non-increasing and non-negative. Chooses p with $\sum_{n>p} u_n < 1$, then derives $u_p \leq u_p \cdot \sum_{n>p} u_n < u_p$, a contradiction unless $u_p = 0$. Hence all u's are zero. Answer: $\boxed{0}$.

Student Solution (abridged)

Sets $n = 0$: $u_0 = \sum_{k=1}^{\infty} u_k^2$. Sets $n = 1$: $u_1 = \sum_{k=2}^{\infty} u_k^2$. Subtracts to get $u_0 = u_1^2 + u_1$, a quadratic. Solves: $u_1 = (-1 + \sqrt{1 + 4u_0})/2$. Claims $u_0 = 1/4$ is possible, giving $u_1 = 1/2$. Answer: $\boxed{1/2}$ (incorrect—the correct answer is 0).

No-CoT Judge (score: 4.5 / 5)

"The model correctly derives the quadratic equation and solves it, finding $u_1 = 1/2$ when $u_0 = 1/4$. The calculations are accurate. So, Intermediate Correctness is 1."

"The model correctly identifies that the maximum occurs at u_1 when u_0 is $1/4$, leading to $u_1 = 1/2$. This directly addresses the task without irrelevant paths. So, Faithfulness is 1."

"The ground truth uses induction and an inequality approach, while the model uses a quadratic equation [...] Both methods aim to find the maximum value, and the model's approach is valid. So, Methodological Alignment is 1."

LC: 1 — FT: 1 — MA: 1 — IC: 1 — EA: 0.5 — **Total: 4.5**

Full-CoT Judge (score: 1.0 / 5)

Scratchpad:
"However, the reasoning doesn't align with the ground truth, which shows all u's are zero. The model's approach is inconsistent because it arrives at a non-zero maximum, contradicting the correct solution. So, Logical Coherence is 0."

"The model's intermediate steps, like solving the quadratic, are mathematically correct. However, the premise that u_0 can be $1/4$ is flawed because the ground truth shows all terms must be zero."

⟨THINKING⟩ summary:
"Logical Coherence (0): The argument is inconsistent. It derives a maximum of $1/2$ but fails to consider that all terms must be zero."

LC: 0 — FT: 0.5 — MA: 0 — IC: 0.5 — EA: 0 — **Total: 1.0**

Figure 5. **The internal-consistency trap (Human: 1.5).** The student solution is internally valid mathematics—the quadratic really does yield $u_1 = 1/2$—but the entire argument rests on the false premise that $u_0 = 1/4$ is achievable. No-CoT evaluates the solution as a standalone argument ("the calculations are accurate"; "the model's approach is valid") without confronting the ground truth's proof that all terms are zero. Full-CoT's scratchpad immediately cross-references: "*doesn't align with the ground truth, which shows all u's are zero*," then the <THINKING> phase assigns consequences. **This example illustrates the core failure: No-CoT checks whether the student's math is self-consistent, not whether its premises are correct.**

Problem (Putnam #62)

Let $f(x)$ be a cubic polynomial with roots x_1, x_2, x_3. Assume that $f(2x)$ is divisible by $f'(x)$. Compute the ratios $x_1 : x_2 : x_3$.

Ground Truth (abridged)

Writes $f(2x) = f'(x)(px + q)$, compares coefficients to get $b = 4a^2/3$, $c = -16a^3/27$. Factors $f(x)$ to find roots $1, -2+2\sqrt{3}\,i, -2-2\sqrt{3}\,i$ (up to scaling by a). Ratios involve **complex numbers**.

Student Solution (abridged)

Notes that roots of $f'(x)$, when doubled, must be roots of $f(x)$. Uses Vieta's formulas. Assumes all roots are real, concludes $x_1 : x_2 : x_3 = 2 : 1 : 1$ without justification. Does not consider complex roots.

No-CoT Judge (score: 3.5 / 5)

"The MODEL_OUTPUT's steps are locally valid and globally consistent. It outlines a logical progression [...] Logical Coherence: 1."

"The approach is compatible with the canonical method. Both the model and ground truth involve setting up integrals [sic] and solving for c [sic]. Methodological Alignment: 1."

"The model concludes the ratio is 2:1:1. However, the ground truth shows the roots are $1, -2+2\sqrt{3}\,i, -2-2\sqrt{3}\,i$, which don't align [...] The model's intermediate step about the ratio might be incorrect, so I'll give 0.5."

LC: 1 — FT: 1 — MA: 1 — IC: 0.5 — EA: 0 — **Total: 3.5**

Full-CoT Judge (score: 1.5 / 5)

Scratchpad:
"However, the model uses Vieta's formulas [...] but it doesn't apply them as thoroughly as the ground truth. The ground truth sets up equations by comparing coefficients, which is a more precise approach [...] 0.5."

⟨THINKING⟩ summary:
"Logical Coherence (0.5): [...] the leap to the ratio 2:1:1 is abrupt and lacks detailed justification."
"Faithfulness (0.5): [...] oversimplifies by assuming real roots and a specific ratio without considering complex roots as in the ground truth."
"Methodological Alignment (0.5): Uses Vieta's formulas [...] but applies it less rigorously than the ground truth."

LC: 0.5 — FT: 0.5 — MA: 0.5 — IC: 0 — EA: 0 — **Total: 1.5**

Figure 6. **Failure to propagate error backward through rubric criteria (Human: 1.5).** The No-CoT judge evaluates each criterion independently: "Does the reasoning flow logically?" (yes, on the surface)—and awards 1. It even *notices* the wrong answer ("the ground truth shows the roots are $1, -2+2\sqrt{3}\,i, \ldots$ which don't align") but confines the penalty to Intermediate Correctness (0.5) without revising the upstream criteria. Full-CoT recognizes that the wrong answer retroactively undermines Logical Coherence ("the leap to 2:1:1 is abrupt"), Faithfulness ("oversimplifies by assuming real roots"), and Methodology ("applies it less rigorously")—propagating the error backward to produce a correctly calibrated total. Note also that No-CoT *hallucinates* problem content: it refers to "setting up integrals and solving for *c*," neither of which appear in this problem.

I. Additional Strategy-Selection Headroom Results

This appendix reports robustness analyses for the MSE-aligned strategy-selection headroom results in Section 4. We include only analyses that answer distinct questions beyond the main table: sensitivity to the stability penalty λ, sensitivity to the token-cost penalty γ, and the distribution of oracle-selected strategies.

I.1. Sensitivity to stability penalty

We first sweep the stability penalty λ while fixing $\gamma = 0$. Across both benchmarks, oracle selection continues to improve over the validation-selected best fixed strategy across all tested λ values, indicating that adaptive headroom is not tied to a single stability weighting.

Table 9. Putnam-AXIOM-Grading sensitivity to the stability penalty λ with $\gamma = 0$. Δ columns report utility improvement under the corresponding λ. MSE is measured on the native 0–5 Putnam scale.

λ	Model	Best fixed	Δ vs fixed	Δ vs comp.	Oracle MSE	Fixed MSE	Comp. MSE
0	DeepSeek-R1-32B	Comparative	0.036	0.036	0.328	1.225	1.225
0	Llama-3.1-70B	Bullet-point	0.030	0.045	0.223	0.971	1.340
0	Qwen2.5-32B	Comparative	0.039	0.039	0.217	1.197	1.197
0	Qwen2.5-72B	Comparative	0.038	0.038	0.175	1.115	1.115
0.25	DeepSeek-R1-32B	Comparative	0.036	0.036	0.340	1.225	1.225
0.25	Llama-3.1-70B	Bullet-point	0.030	0.044	0.230	0.971	1.340
0.25	Qwen2.5-32B	Comparative	0.038	0.038	0.234	1.197	1.197
0.25	Qwen2.5-72B	Comparative	0.037	0.037	0.181	1.115	1.115
0.5	DeepSeek-R1-32B	Comparative	0.037	0.037	0.360	1.225	1.225
0.5	Llama-3.1-70B	Bullet-point	0.031	0.045	0.239	0.971	1.340
0.5	Qwen2.5-32B	Comparative	0.038	0.038	0.251	1.197	1.197
0.5	Qwen2.5-72B	Comparative	0.037	0.037	0.194	1.115	1.115
1	DeepSeek-R1-32B	Comparative	0.041	0.041	0.392	1.225	1.225
1	Llama-3.1-70B	Bullet-point	0.033	0.046	0.264	0.971	1.340
1	Qwen2.5-32B	Comparative	0.039	0.039	0.288	1.197	1.197
1	Qwen2.5-72B	Comparative	0.037	0.037	0.221	1.115	1.115
2	DeepSeek-R1-32B	Comparative	0.050	0.050	0.469	1.225	1.225
2	Llama-3.1-70B	Bullet-point	0.038	0.049	0.323	0.971	1.340
2	Qwen2.5-32B	No-CoT	0.032	0.042	0.349	1.068	1.197
2	Qwen2.5-72B	No-CoT	0.035	0.039	0.268	1.088	1.115

Table 10. IMO-GradingBench sensitivity to the stability penalty λ with $\gamma = 0$. Δ columns report utility improvement under the corresponding λ. MSE is measured on the native 0–7 IMO scale.

λ	**Model**	**Best fixed**	**Δ vs fixed**	**Δ vs comp.**	**Oracle MSE**	**Fixed MSE**	**Comp. MSE**
0	DeepSeek-R1-32B	Debate	0.048	0.079	11.336	13.672	15.217
0	Llama-3.1-70B	Comparative	0.036	0.036	6.584	8.367	8.367
0	Qwen2.5-32B	Comparative	0.067	0.067	7.878	11.185	11.185
0	Qwen2.5-72B	Debate	0.026	0.094	9.961	11.233	14.559
0.25	DeepSeek-R1-32B	Debate	0.048	0.079	11.336	13.672	15.217
0.25	Llama-3.1-70B	Comparative	0.037	0.037	6.586	8.367	8.367
0.25	Qwen2.5-32B	Comparative	0.067	0.067	7.878	11.185	11.185
0.25	Qwen2.5-72B	Debate	0.027	0.093	9.962	11.233	14.559
0.5	DeepSeek-R1-32B	Debate	0.049	0.078	11.340	13.672	15.217
0.5	Llama-3.1-70B	Comparative	0.039	0.039	6.592	8.367	8.367
0.5	Qwen2.5-32B	Comparative	0.067	0.067	7.878	11.185	11.185
0.5	Qwen2.5-72B	Debate	0.028	0.093	9.966	11.233	14.559
1	DeepSeek-R1-32B	Debate	0.051	0.077	11.352	13.672	15.217
1	Llama-3.1-70B	Comparative	0.041	0.041	6.633	8.367	8.367
1	Qwen2.5-32B	Comparative	0.067	0.067	7.878	11.185	11.185
1	Qwen2.5-72B	Debate	0.029	0.093	9.968	11.233	14.559
2	DeepSeek-R1-32B	Debate	0.055	0.076	11.416	13.672	15.217
2	Llama-3.1-70B	Comparative	0.048	0.048	6.779	8.367	8.367
2	Qwen2.5-32B	Comparative	0.067	0.067	7.878	11.185	11.185
2	Qwen2.5-72B	Debate	0.033	0.092	9.975	11.233	14.559

I.2. Sensitivity to token-cost penalty

We next sweep the token-cost penalty γ while fixing $\lambda = 1$. Gains shrink as token cost is penalized more heavily, but remain positive across all tested settings, indicating that the headroom is not simply an artifact of selecting expensive multi-call strategies.

Table 11. Putnam-AXIOM-Grading sensitivity to the cost penalty γ with $\lambda = 1$. Δ columns report utility improvement under the corresponding γ; token counts are reconstructed approximate token counts.

γ	**Model**	**Best fixed**	**Δ vs fixed**	**Δ vs comp.**	**Oracle tokens**	**Fixed tokens**	**Comp. tokens**
0	DeepSeek-R1-32B	Comparative	0.041	0.041	4775	2582	2582
0	Llama-3.1-70B	Bullet-point	0.033	0.046	3514	1956	2057
0	Qwen2.5-32B	Comparative	0.039	0.039	3362	2121	2121
0	Qwen2.5-72B	Comparative	0.037	0.037	3830	2118	2118
0.01	DeepSeek-R1-32B	Comparative	0.039	0.039	3009	2582	2582
0.01	Llama-3.1-70B	Bullet-point	0.031	0.044	2300	1956	2057
0.01	Qwen2.5-32B	Comparative	0.037	0.037	2397	2121	2121
0.01	Qwen2.5-72B	No-CoT	0.031	0.034	2822	1677	2118
0.05	DeepSeek-R1-32B	Comparative	0.038	0.038	2448	2582	2582
0.05	Llama-3.1-70B	Bullet-point	0.030	0.044	2063	1956	2057
0.05	Qwen2.5-32B	No-CoT	0.024	0.037	1977	1680	2121
0.05	Qwen2.5-72B	No-CoT	0.023	0.031	2214	1677	2118
0.1	DeepSeek-R1-32B	Brief-CoT	0.035	0.040	2364	2063	2582
0.1	Llama-3.1-70B	Bullet-point	0.029	0.044	1952	1956	2057
0.1	Qwen2.5-32B	No-CoT	0.021	0.040	1858	1680	2121
0.1	Qwen2.5-72B	No-CoT	0.019	0.032	1946	1677	2118
0.25	DeepSeek-R1-32B	Brief-CoT	0.028	0.049	2219	2063	2582
0.25	Llama-3.1-70B	Bullet-point	0.032	0.050	1832	1956	2057
0.25	Qwen2.5-32B	No-CoT	0.016	0.052	1756	1680	2121
0.25	Qwen2.5-72B	No-CoT	0.014	0.043	1753	1677	2118

Table 12. IMO-GradingBench sensitivity to the cost penalty γ with $\lambda = 1$. Δ columns report utility improvement under the corresponding γ; token counts are reconstructed approximate token counts.

γ	**Model**	**Best fixed**	**Δ vs fixed**	**Δ vs comp.**	**Oracle tokens**	**Fixed tokens**	**Comp. tokens**
0	DeepSeek-R1-32B	Debate	0.051	0.077	7278	16947	4705
0	Llama-3.1-70B	Comparative	0.041	0.041	6099	4705	4705
0	Qwen2.5-32B	Comparative	0.067	0.067	5448	4705	4705
0	Qwen2.5-72B	Debate	0.029	0.093	10569	15509	4705
0.01	DeepSeek-R1-32B	Debate	0.061	0.075	6820	16947	4705
0.01	Llama-3.1-70B	Comparative	0.040	0.040	5394	4705	4705
0.01	Qwen2.5-32B	Comparative	0.067	0.067	5403	4705	4705
0.01	Qwen2.5-72B	Debate	0.035	0.086	10256	15509	4705
0.05	DeepSeek-R1-32B	Bullet-point	0.052	0.068	6011	4700	4705
0.05	Llama-3.1-70B	Comparative	0.038	0.038	4929	4705	4705
0.05	Qwen2.5-32B	Comparative	0.064	0.064	5087	4705	4705
0.05	Qwen2.5-72B	Comparative	0.064	0.064	8464	4705	4705
0.1	DeepSeek-R1-32B	Bullet-point	0.047	0.063	5353	4700	4705
0.1	Llama-3.1-70B	Comparative	0.037	0.037	4763	4705	4705
0.1	Qwen2.5-32B	Comparative	0.062	0.062	4982	4705	4705
0.1	Qwen2.5-72B	Comparative	0.046	0.046	7043	4705	4705
0.25	DeepSeek-R1-32B	Bullet-point	0.044	0.059	4759	4700	4705
0.25	Llama-3.1-70B	Comparative	0.037	0.037	4694	4705	4705
0.25	Qwen2.5-32B	Comparative	0.059	0.059	4777	4705	4705
0.25	Qwen2.5-72B	Comparative	0.024	0.024	5113	4705	4705

I.3. Oracle strategy frequencies

The oracle does not collapse to a single strategy. Putnam selections are distributed across nearly all candidate strategies for each model; IMO selections are more concentrated but remain strongly model-dependent. This supports the interpretation that strategy utility is instance-conditional rather than captured by a single global ranking.

Table 13. Putnam-AXIOM-Grading oracle-selected strategy counts under $\lambda = 1, \gamma = 0$. Counts are over held-out test instances.

Model	Brief	Bullet	Comp.	CoVe	Debate	Full	G-Eval	Multi	No-CoT	Overseer	Quote	Self	Struct.
DeepSeek-R1-32B	26	38	42	21	21	24	23	29	25	25	33	20	45
Llama-3.1-70B	26	60	41	21	24	34	23	20	43	15	21	21	35
Qwen2.5-32B	35	50	35	38	25	20	6	18	83	12	10	21	31
Qwen2.5-72B	39	26	48	17	63	21	25	11	64	12	15	13	28

Table 14. IMO-GradingBench oracle-selected strategy counts under $\lambda = 1, \gamma = 0$. Counts are over held-out test instances.

Model	Brief	Bullet	Comp.	CoVe	Debate	Full	G-Eval	Multi	No-CoT	Overseer	Quote	Self	Struct.
DeepSeek-R1-32B	97	107	64	11	114	36	37	7	205	10	19	34	47
Llama-3.1-70B	26	68	296	11	35	16	7	32	255	4	10	8	20
Qwen2.5-32B	134	53	130	15	21	28	22	5	204	0	7	65	104
Qwen2.5-72B	43	19	81	1	403	2	1	1	207	0	2	7	21

J. Prompts

J.1. Putnam-AXIOM-Grading Prompts

Base Context (Common to All Strategies)

Note: All strategies typically begin with this context block.
<CONTEXT> [Rubric Text Omitted] </CONTEXT>
<PROBLEM> {problem_statement} </PROBLEM>
<GROUND_TRUTH_SOLUTION> {ground_truth_solution} </GROUND_TRUTH_SOLUTION>
<MODEL_OUTPUT> {model_output} </MODEL_OUTPUT>

SINGLE-TURN STRATEGIES

Strategy: No CoT

<INSTRUCTIONS> Provide your scores DIRECTLY in the specified JSON format inside <json> tags. Do NOT include any reasoning.
Your response MUST include a <json>...</json> block and NOTHING after it.
Here is an example response: <json> { "score_logical_coherence": 1, "score_faithfulness_to_task": 1, "score_methodological_alignment": 1, "score_intermediate_correctness": 1, "score_error_awareness": 1, "score_total": 5 } </json> </INSTRUCTIONS>

Strategy: Brief CoT

<INSTRUCTIONS> First, write ONLY 1-2 SHORT sentences inside <THINKING> tags summarizing your overall assessment. Do NOT explain your entire process. Just state your assessment directly. Then provide your scores in the specified JSON format inside <json> tags.
Example format: <THINKING> The solution correctly applies integration by parts but makes an arithmetic error in step 3. Otherwise reasoning is sound. </THINKING> <json> { "score_logical_coherence": 1, "score_faithfulness_to_task": 1, "score_methodological_alignment": 1, "score_intermediate_correctness": 0, "score_error_awareness": 0, "score_total": 3 } </json> </INSTRUCTIONS>

Strategy: Structured CoT

<INSTRUCTIONS> You MUST follow this exact three-step structure inside <THINKING> tags:
STEP 1 - SUMMARIZE: In 2-3 sentences, summarize what the model's solution attempts to do.
STEP 2 - CHECK KEY STEPS: List the 2-4 most critical steps/calculations. For each, note if it is correct (✓) or incorrect (×).
STEP 3 - DECIDE SCORES: Based on your analysis, determine each rubric score with one-line justification.
Then provide your final scores in <json> tags.
Example: <THINKING> STEP 1 - SUMMARIZE: The model attempts to solve the integral using substitution, converting the expression and evaluating bounds.
STEP 2 - CHECK KEY STEPS: - Substitution $u = x^2 + 1$: ✓Valid transformation - Computing $du = 2xdx$: ✓Correct derivative - Final evaluation at bounds: ×Arithmetic error (should be 15, got 12)
STEP 3 - DECIDE SCORES: - Logical coherence: 1 (steps follow logically) - Faithfulness: 1 (addresses the problem) - Methodological alignment: 1 (standard approach) - Intermediate correctness: 0 (arithmetic error) - Error awareness: 0 (error not noticed) </THINKING> <json> { "score_logical_coherence": 1, "score_faithfulness_to_task": 1, "score_methodological_alignment": 1, "score_intermediate_correctness": 0, "score_error_awareness": 0, "score_total": 3 } </json> </INSTRUCTIONS>

Strategy: Full CoT

<INSTRUCTIONS> You MUST output your response in this EXACT format: 1. FIRST, write your step-by-step reasoning inside <THINKING>...</THINKING> tags 2. THEN, output the scores inside <json>...</json> tags

For instance, a valid grading response to a correct answer could look like this: "<THINKING> Logical Coherence (1): The argument proceeds in a clear sequence, with each step following from the last; no leaps or contradictions detected. Faithfulness to Task (1): The response stays focused on the stated goal and uses only information relevant to answering the problem. The response never strays away from the given task. Methodological Alignment (1): The chosen method is standard for this type of problem and is applied appropriately. Intermediate Correctness (1): Intermediate claims match known identities/definitions and are used consistently. Error Awareness / Self-Correction (1): No mistakes observed; scope conditions are implicitly respected and edge cases would not change the conclusion. </THINKING> <json> { "score_logical_coherence": 1, "score_faithfulness_to_task": 1, "score_methodological_alignment": 1, "score_intermediate_correctness": 1, "score_error_awareness": 1, "score_total": 5 } </json>"

A valid grading response to an incorrect answer could look like this: "<THINKING> Logical Coherence (0): The reasoning is inconsistent: it mixes unrelated steps, reverses implications, and reaches conclusions that don't follow from prior claims. Faithfulness to Task (1): The response attempts to answer the stated question and references the required quantities, but stays narrowly focused while still arriving at the wrong result. Methodological Alignment (0): The chosen approach conflicts with standard methods for this problem and misapplies key definitions/identities. Intermediate Correctness (0): Several intermediate statements are false (e.g., misuse of a determinant identity and an invalid simplification), so later conclusions are unsupported. Error Awareness / Self-Correction (0): No checks, sanity tests, or alternative cases are considered; contradictions and edge conditions go unaddressed. <json> { "score_logical_coherence": 0, "score_faithfulness_to_task": 1, "score_methodological_alignment": 0, "score_intermediate_correctness": 0, "score_error_awareness": 1, "score_total": 1 } </json>" </INSTRUCTIONS>

Strategy: Self-Critique

<INSTRUCTIONS> Follow this two-phase evaluation inside <THINKING> tags:

PHASE 1 - INITIAL ASSESSMENT: Quickly evaluate the solution and propose initial scores.

PHASE 2 - SELF-CRITIQUE: Challenge your initial assessment. Ask: "Am I being too harsh or too lenient?" "Did I miss anything?" Adjust scores if needed.

Then provide your final scores in <json> tags.

Example: <THINKING> PHASE 1 - INITIAL ASSESSMENT: The solution uses the correct method but has a sign error. Initial scores: coherence=1, faithfulness=1, method=1, intermediate=0, awareness=0.

PHASE 2 - SELF-CRITIQUE: Wait - I should check if the sign error propagates. Looking again... yes, it affects the final answer but the method is still valid. The model did attempt to verify the answer at the end, showing some error awareness even if unsuccessful. Adjusting error_awareness to 0.5. </THINKING> <json> { "score_logical_coherence": 1, "score_faithfulness_to_task": 1, "score_methodological_alignment": 1, "score_intermediate_correctness": 0, "score_error_awareness": 0.5, "score_total": 3.5 } </json> </INSTRUCTIONS>

Strategy: Bullet Point

<INSTRUCTIONS> Use bullet points inside <THINKING> tags to systematically evaluate each criterion:

• Logical Coherence: [observation] → [score] • Faithfulness to Task: [observation] → [score] • Methodological Alignment: [observation] → [score] • Intermediate Correctness: [observation] → [score] • Error Awareness: [observation] → [score] • Total: [sum]

Then provide scores in <json> tags.

Example: <THINKING> • Logical Coherence: Steps follow clear sequence with proper transitions → 1 • Faithfulness to Task: Directly addresses the integral as asked → 1 • Methodological Alignment: Uses substitution, same as ground truth → 1 • Intermediate Correctness: Calculation error in step 4 (8x3=24, not 26) → 0 • Error Awareness: No verification attempted → 0 • Total: 3 </THINKING> <json> { "score_logical_coherence": 1, "score_faithfulness_to_task": 1, "score_methodological_alignment": 1, "score_intermediate_correctness": 0, "score_error_awareness": 0, "score_total": 3 } </json> </INSTRUCTIONS>

Strategy: Comparative

<INSTRUCTIONS> Inside <THINKING> tags, perform a comparative analysis:
1. KEY APPROACH COMPARISON: - Ground Truth approach: [brief description] - Model approach: [brief description] - Match? [Yes/No/Partial]
2. CRITICAL DIFFERENCES: List any significant differences in reasoning or results.
3. ERROR ANALYSIS: Note any errors in the model output that differ from ground truth.
4. SCORING RATIONALE: Brief justification for each score based on the comparison.
Then provide scores in <json> tags.
Example: <THINKING> 1. KEY APPROACH COMPARISON: - Ground Truth: Uses integration by parts twice - Model: Uses integration by parts twice - Match? Yes
2. CRITICAL DIFFERENCES: - Model skips showing the second application explicitly - Final constant differs (model has +C, GT shows specific value)
3. ERROR ANALYSIS: - Sign error in step 2: -cos(x) written as cos(x) - This propagates to incorrect final answer
4. SCORING RATIONALE: - Coherence: 1 (logical flow maintained) - Faithfulness: 1 (addresses the integral) - Method: 1 (same technique) - Intermediate: 0 (sign error) - Awareness: 0 (not caught) </THINKING> <json> { "score_logical_coherence": 1, "score_faithfulness_to_task": 1, "score_methodological_alignment": 1, "score_intermediate_correctness": 0, "score_error_awareness": 0, "score_total": 3 } </json> </INSTRUCTIONS>

Strategy: Quote Forcing

<INSTRUCTIONS> You MUST output your response in this EXACT format: 1. FIRST, write your reasoning inside <THINKING>...</THINKING> tags, citing evidence from <MODEL_OUTPUT> for EVERY evaluative claim with short direct quotes like: "..." 2. THEN, output the scores inside <json>...</json> tags
Your response MUST start with <THINKING> and include quotes before the JSON block. Your response MUST include a <json>...</json> block and NOTHING after it.
Example: <THINKING> The model claims to derive the formula, stating "$f(x) = x^2 + 2x$" [Quote from Line 4]. However, this contradicts the prompt which asked for "$f(x) = x^3$". The model later attempts to correct this: "Wait, I meant cubed" [Quote from Line 6], but the calculation remains quadratic. </THINKING> <json> { "score_logical_coherence": 0, "score_faithfulness_to_task": 1, "score_methodological_alignment": 0, "score_intermediate_correctness": 0, "score_error_awareness": 0, "score_total": 1 } </json> </INSTRUCTIONS>

Multi-Turn Strategies

Strategy: Debate

Advocate Judge (Pro): <INSTRUCTIONS> You are the "Advocate Judge". Your goal is to argue why this response deserves the HIGHEST POSSIBLE SCORE. Highlight every correct step, and every logical connection. Output your argument in <ARGUMENT_PRO> tags. </INSTRUCTIONS>

Critic Judge (Con): <INSTRUCTIONS> You are the "Critic Judge". Your goal is to argue why this response deserves the LOWEST POSSIBLE SCORE. Scrutinize every step, look for any logical gap, and interpret ambiguity as failure. Output your argument in <ARGUMENT_CON> tags. </INSTRUCTIONS>

Arbiter: <ARGUMENT_PRO> {pro_arg} </ARGUMENT_PRO>
<ARGUMENT_CON> {con_arg} </ARGUMENT_CON>
<INSTRUCTIONS> You are the Arbiter. Read the Pro and Con arguments. Decide which is more reasonable based on the Rubric. Usually the truth lies in the middle, but be decisive if one side is clearly hallucinating. Provide your

neutral reasoning in <THINKING> and final scores in <json>. </INSTRUCTIONS>

Strategy: G-Eval

Step 1 (Plan): <INSTRUCTIONS> Your task is to convert the Rubric into atomic evaluation steps specific to this problem. 1. Break down the specific requirements of the problem and ground truth. 2. Create a checklist of 3-5 atomic steps/checks. 3. Mark each step as [CRITICAL] or [NON-CRITICAL].
Output inside <CHECKLIST_PLAN> tags.
Example: <CHECKLIST_PLAN> 1. [CRITICAL] Does the model correctly identify the integration bounds (0 to infinity)? 2. [CRITICAL] Is the substitution $u = x^2$ applied correctly? 3. [NON-CRITICAL] Is the final answer simplified to the simplest form? 4. [CRITICAL] Does the model check for convergence? </CHECKLIST_PLAN> </INSTRUCTIONS>

Step 2 (Score): <EVALUATION_PLAN> {step1_output} </EVALUATION_PLAN>
<INSTRUCTIONS> Evaluate the model output against the Checklist Plan above. For each criterion in the original rubric, calculate the score using this logic: - Score = 1 if ALL [CRITICAL] checks pass AND ¡= 1 [NON-CRITICAL] check fails. - Score = 0 if ANY [CRITICAL] check fails. - Score = 0.5 otherwise.
Show your work in <THINKING> tags, verifying each checklist item as PASS/FAIL. Output final scores in <json>. </INSTRUCTIONS>

Strategy: CoVe

Step 2 (Plan Verification): <DRAFT_EVALUATION> {draft_response} </DRAFT_EVALUATION>
<INSTRUCTIONS> Based on the draft evaluation above and the problem, generate 3-5 specific ”Verification Questions” to fact-check the draft’s claims. Focus on checking if the draft missed errors or hallucinations in the model output. Output inside <VERIFICATION_QUESTIONS>.
Example: <VERIFICATION_QUESTIONS> - Did the model actually calculate the derivative of log(x) correctly in step 2? - The draft claims the model followed the prompt constraints; does the model output contain the forbidden word ”delve”? - Is the final constant matching the Ground Truth exactly? </VERIFICATION_QUESTIONS> </INSTRUCTIONS>

Step 3 (Execute Verification): <QUESTIONS> {questions} </QUESTIONS>
<INSTRUCTIONS> Answer the verification questions above objectively based *strictly* on the Model Output and Ground Truth. Do not grade yet, just answer the questions inside <VERIFICATION_ANSWERS>.
Example: <VERIFICATION_ANSWERS> 1. No, the model calculated $1/x^2$ instead of $1/x$. 2. Yes, the model used the word ”delve”. 3. No, Ground Truth is 15, Model is 14. </VERIFICATION_ANSWERS> </INSTRUCTIONS>

Step 4 (Finalize): <DRAFT_EVALUATION> {draft} </DRAFT_EVALUATION>
<VERIFICATION_RESULTS> {qa_pairs} </VERIFICATION_RESULTS>
<INSTRUCTIONS> Review the Draft Evaluation in light of the Verification Results. If the verification found errors in the model that the draft missed, downgrade the score. If the verification confirms the draft was correct, maintain the score. Provide final reasoning and <json>. </INSTRUCTIONS>

Strategy: Multi-Judge (Base Judges)

Note: Three separate judges grade the problem in parallel, each with a distinct persona prepended to the standard Full CoT instructions.
Judge 1 (Pedantic): ”You are a strict, pedantic judge who focuses on details.”
Judge 2 (Holistic): ”You are a holistic judge who focuses on the big picture.”
Judge 3 (Teacherly): ”You are a helpful teacher grading a student.”

All three judges then receive the standard Full CoT prompt: <INSTRUCTIONS> You MUST output your response in this EXACT format: 1. FIRST, write your step-by-step reasoning inside <THINKING>...</THINKING> tags 2. THEN, output the scores inside <json>...</json> tags

For instance, a valid grading response to a correct answer could look like this: "<THINKING> ... </THINKING> <json> { "score_logical_coherence": 1, ... } </json>" </INSTRUCTIONS>

Strategy: Multi-Judge (Chair Judge)

<BASE_JUDGMENTS> <JUDGE_1> ... </JUDGE_1> <JUDGE_2> ... </JUDGE_2> <JUDGE_3> ... </JUDGE_3> </BASE_JUDGMENTS>
<INSTRUCTIONS> You are the Chair Judge. You have received evaluations from 3 other judges. Synthesize their insights. If they disagree, look at the Ground Truth to break the tie. Provide a final, consolidated evaluation in <THINKING> and scores in <json>. </INSTRUCTIONS>

Strategy: Overseer Audit

<INSTRUCTIONS> Assess whether the JUDGE_OUTPUT contains any logical errors in its evaluation for EACH rubric component. If there is NO thinking chain, mark the entire evaluation as problematic.
For each component, answer if there is a logical error (true/false) and briefly explain why. Errors include: misapplication of the rubric, inconsistent or contradictory reasoning, unsupported claims, overlooking critical evidence from PROBLEM/GT/MODEL_OUTPUT, or internally inconsistent scoring rationale.
Return STRICT JSON inside <json> tags, with this exact schema and keys: { "logical_coherence": "...", "logical_coherence_verdict": ¡true—false¿, "faithfulness_to_task": "...", "faithfulness_to_task_verdict": ¡true—false¿, "methodological_alignment": "...", "methodological_alignment_verdict": ¡true—false¿, "intermediate_correctness": "...", "intermediate_correctness_verdict": ¡true—false¿, "error_awareness": "...", "error_awareness_verdict": ¡true—false¿, "overall_error": ¡true—false¿, "notes": "¡optional notes¿" } Only output the JSON within <json>...</json> tags. </INSTRUCTIONS>
Rubric the judge was supposed to apply: {RUBRIC_TEXT}
<PROBLEM> {problem["problem_statement"]} </PROBLEM>
<GROUND_TRUTH_SOLUTION> {problem["ground_truth_solution"]} </GROUND_TRUTH_SOLUTION>
<MODEL_OUTPUT> {problem["model_output"]} </MODEL_OUTPUT>
<JUDGE_OUTPUT> {judge_text} </JUDGE_OUTPUT>

J.2. IMO-GradingBench Prompts

Base Context (Common to All Strategies)

Note: All strategies begin with this context, generated by `_base_content(p)`.
<RUBRIC> You are an expert Math Olympiad grader. Grade the student's solution on the standard IMO 0-7 scale: 7: Completely correct solution. 6: Correct solution with very minor slips (e.g., a sign error that doesn't affect the logic). 5: Main theorem/logic is established, but a significant case or detail is missed. 4: Significant progress towards the solution (e.g., reducing the problem to a known solved state), but main proof incomplete. 3: Non-trivial partial results that are steps towards the solution. 2: Some relevant formulas or observations, but no significant progress. 1: Fundamental misunderstanding or only trivial observations. 0: Completely incorrect or blank.
Output your score as an integer from 0 to 7. </RUBRIC>
<PROBLEM> {problem['problem']} </PROBLEM>
<GROUND_TRUTH> {problem['solution']} </GROUND_TRUTH>
<STUDENT_ANSWER> {problem['model_output']} </STUDENT_ANSWER>

SINGLE-TURN STRATEGIES

Strategy: No CoT

<INSTRUCTIONS> Grade the answer 0-7. Output JSON: <json>{'score': X}</json> Your response MUST include a <json>...</json> block and nothing after it.
Examples (follow the formatting style shown): Good-solution-style example: <json>{'score': 7}</json> Bad-solution-style example: <json>{'score': 1}</json> </INSTRUCTIONS>

Strategy: Brief CoT

<INSTRUCTIONS> Write ONLY 1-2 short sentences in <THINKING> tags summarizing the evaluation, then output <json>{'score': X}</json>. Your response MUST include a <json>...</json> block and nothing after it.
Example: <THINKING> The student follows the main idea but misses a critical case, so the proof is incomplete. </THINKING> <json>{'score': 5}</json> </INSTRUCTIONS>

Strategy: Structured CoT

<INSTRUCTIONS> You MUST follow this exact structure in <THINKING>:
STEP 1 - SUMMARIZE (2-3 sentences): What is the student trying to do? STEP 2 - CHECK KEY STEPS (2-4 bullets): For each critical step, mark ✓/× with a brief reason. STEP 3 - DECIDE SCORE (1-2 sentences): Map gaps/errors to the IMO 0-7 rubric and decide the integer score.
Then output <json>{'score': X}</json>. Your response MUST include a <json>...</json> block and nothing after it.
Example: <THINKING> STEP 1 - SUMMARIZE: The student attempts to prove the statement by establishing a lemma and applying it to a case split.
STEP 2 - CHECK KEY STEPS: - Lemma statement and proof: ✓ correct and complete - Case split covers all possibilities: × misses the equality/boundary case - Final conclusion matches the prompt: ✓
STEP 3 - DECIDE SCORE: Main approach is correct but a significant case is missing, so substantial partial credit but not full. </THINKING> <json>{'score': 5}</json> </INSTRUCTIONS>

Strategy: Full CoT

<INSTRUCTIONS> You MUST output your response in this exact format: 1. FIRST, write your step-by-step reasoning inside <THINKING>...</THINKING> tags 2. THEN, output the score inside <json>{'score': X}</json> Your response MUST start with <THINKING> and include both the reasoning and the JSON block.
Examples (follow the formatting and tone): Good-solution-style example: <THINKING> The student correctly identifies the key move: they rewrite the target inequality into a symmetric form and then apply Cauchy–Schwarz (in Engel/Titu's lemma form) to bound the main expression. They explicitly justify why the substitution/normalization is valid (e.g., they note the variables are positive so denominators are nonzero), and they track the equality case by checking when the C–S step is tight. After the C–S bound, they close the argument with a clean AM–GM step to match the exact constant in the ground truth, and they verify that the derived equality conditions match the problem's constraints. No required case is omitted, and the final conclusion is stated in the exact form requested. </THINKING> <json>{'score': 7}</json>
Bad-solution-style example: <THINKING> The student attempts to mimic the official approach by invoking Cauchy–Schwarz, but they apply it to the wrong quantities (they bound a different expression than the one appearing in the problem), so the resulting inequality does not imply the target statement. They also divide by an expression that can be zero without checking the domain, so a critical step is not logically valid for all inputs. Even if one overlooks that gap, the work never addresses the equality case (the ground truth requires identifying when equality holds), and the final line claims "therefore the minimum is achieved" without proving existence. Overall there are relevant ideas, but the core implication is unsupported and at least one necessary condition/case is missing. </THINKING> <json>{'score': 2}</json> </INSTRUCTIONS>

Strategy: Self-Critique

<INSTRUCTIONS> You MUST output your response in this exact format: 1. FIRST, write your analysis inside <THINKING>...</THINKING> tags with two phases: - PHASE 1 - INITIAL ASSESSMENT: Evaluate and propose a tentative IMO 0-7 score. - PHASE 2 - SELF-CRITIQUE: Challenge your own assessment (too harsh/lenient? missed a case? over-trusting an unproven claim?). Adjust if needed. 2. THEN, output the final score inside <json>{'score': X}</json>
Your response MUST start with <THINKING> and include both phases before the JSON block.
Example: <THINKING> PHASE 1 - INITIAL ASSESSMENT: The student has the right main idea and proves a key lemma, but the last step is not justified; tentative score: 4.
PHASE 2 - SELF-CRITIQUE: On review, the last step implicitly assumes a condition that was never established, so the argument does not actually reach the main claim. However, the partial results are still substantial. Keeping 4. </THINKING> <json>{'score': 4}</json> </INSTRUCTIONS>

Strategy: Bullet Point

<INSTRUCTIONS> Use bullet points in <THINKING> to justify an IMO 0-7 score: • Correctness of main idea • Handling of key lemmas/claims • Case coverage / edge cases • Logical gaps / unjustified leaps • Final conclusion vs problem requirement Then output <json>{'score': X}</json>. Your response MUST include a <json>...</json> block and nothing after it.
Example: <THINKING> • Main idea: correct direction and setup • Key lemma: stated but only partially justified • Case coverage: misses one required case • Logical gaps: one nontrivial implication is asserted • Final conclusion: matches statement but proof incomplete </THINKING> <json>{'score': 5}</json> </INSTRUCTIONS>

Strategy: Comparative

<INSTRUCTIONS> You MUST output your response in this exact format: 1. FIRST, write your comparison inside <THINKING>...</THINKING> tags covering: a) Approach match? (Yes/Partial/No) b) Key differences (2-4 bullets) c) Any incorrect claims or missing cases d) Decide IMO 0-7 score from that comparison 2. THEN, output the score inside <json>{'score': X}</json>
Your response MUST start with <THINKING> and include the full comparison before the JSON block.
Example: <THINKING> 1) Approach match? Partial 2) Differences: - Student proves an easier sub-claim than required - Ground truth uses a stronger lemma the student never establishes 3) Missing case: boundary case not addressed 4) Score: main progress but incomplete proof </THINKING> <json>{'score': 4}</json> </INSTRUCTIONS>

Strategy: Quote Forcing

<INSTRUCTIONS> You MUST output your response in this exact format: 1. FIRST, write your reasoning inside <THINKING>...</THINKING> tags, citing evidence from <STUDENT_ANSWER> for EVERY evaluative claim with short direct quotes like: "..." 2. THEN, output the score inside <json>{'score': X}</json>
Your response MUST start with <THINKING> and include quotes before the JSON block.
Example: <THINKING> The student claims the key step holds, writing "therefore X implies Y" but never proves the implication; the only support is "clearly" with no argument. They do correctly set up a relevant identity: "Let f(n)=..." which matches the needed setup, but the proof stops after "it follows" without closing the final case. </THINKING> <json>{'score': 3}</json> </INSTRUCTIONS>

Multi-Turn Strategies

Strategy: Debate (Advocate/Critic/Arbiter)

Pro Judge: <INSTRUCTIONS> You are the 'Advocate'. Argue for the HIGHEST possible score. Output argument in <ARGUMENT_PRO> tags.
Examples (follow the formatting style): Good-solution-style example: <ARGUMENT_PRO> The solution mirrors the official approach, validates key steps, and reaches the correct final result, so it deserves a top score. </ARGUMENT_PRO> Bad-solution-style example: <ARGUMENT_PRO> Even if the final answer is wrong, the method closely follows the correct outline and establishes useful intermediate results, so it should receive a nonzero score. </ARGUMENT_PRO> </INSTRUCTIONS>

Con Judge: <INSTRUCTIONS> You are the 'Critic'. Argue for the LOWEST possible score. Output argument in <ARGUMENT_CON> tags.
Examples (follow the formatting style): Good-solution-style example: <ARGUMENT_CON> Despite a mostly correct outline, the solution skips a critical case and does not justify a key implication, so it should not receive full credit. </ARGUMENT_CON> Bad-solution-style example: <ARGUMENT_CON> The work is incoherent, uses invalid steps, and never reaches the required conclusion, so it deserves a very low score. </ARGUMENT_CON> </INSTRUCTIONS>

Arbiter: <ARGUMENT_PRO> {pro_argument} </ARGUMENT_PRO> <ARGUMENT_CON> {con_argument} </ARGUMENT_CON> <INSTRUCTIONS> Review the arguments and decide the score. Output <json>{'score': X}</json> Your response MUST include a <json>...</json> block and nothing after it.
Examples (follow the output format exactly): Good-solution-style example: <json>{'score': 7}</json> Bad-solution-style example: <json>{'score': 1}</json> </INSTRUCTIONS>

Strategy: G-Eval

Step 1: <INSTRUCTIONS> Create a checklist of 3-5 math checks to validate the answer. Output in <CHECKLIST> tags.
Example (follow this structure): <CHECKLIST> 1. Verify the key lemma matches the ground truth. 2. Check the critical case split is handled correctly. 3. Confirm the final numeric value matches the official solution. </CHECKLIST> </INSTRUCTIONS>

Step 2: <CHECKLIST> {checklist} </CHECKLIST> <INSTRUCTIONS> Use the checklist to grade 0-7. Output <json>{'score': X}</json> Your response MUST include a <json>...</json> block and nothing after it.
Examples (follow the output format exactly): Good-solution-style example: <json>{'score': 7}</json> Bad-solution-style example: <json>{'score': 1}</json> </INSTRUCTIONS>

Strategy: CoVe

Step 1 (Draft): <DRAFT> {draft_reasoning} </DRAFT> <INSTRUCTIONS> Generate 3 Yes/No verification questions to check for errors in the draft. Output in <QUESTIONS>.
Example (follow this structure): <QUESTIONS> 1. Did the model correctly prove the key lemma used in the final step? 2. Does the model's final value match the ground truth? 3. Are any required cases omitted? </QUESTIONS> </INSTRUCTIONS>

Step 2 (Answer): <QUESTIONS> {questions} </QUESTIONS> <INSTRUCTIONS> Answer these questions objectively based on the student's work. Output in <ANSWERS>.
Example (follow this structure): <ANSWERS> 1. No. 2. Yes. 3. No. </ANSWERS> </INSTRUCTIONS>

Step 3 (Refine): <DRAFT> {draft_reasoning} </DRAFT> <VERIFICATION> {qa_pairs} </VERIFICATION> <INSTRUCTIONS> Refine the score based on verification. Output <json>{'score': X}</json> Your response MUST include a <json>...</json> block and nothing after it.
Examples (follow the output format exactly): Good-solution-style example: <json>{'score': 7}</json> Bad-solution-style example: <json>{'score': 1}</json> </INSTRUCTIONS>

Strategy: Multi-Judge (Base Judges)

Note: Three separate judges grade the problem in parallel, each with a distinct persona prepended to the standard Full CoT instructions.
Judge 1 (Pedantic): "You are a strict, pedantic judge who focuses on details."
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All three judges then receive the standard Full CoT prompt: <INSTRUCTIONS> You MUST output your response in this exact format: 1. FIRST, write your step-by-step reasoning inside <THINKING>...</THINKING> tags 2. THEN, output the score inside <json>{'score': X}</json> ... </INSTRUCTIONS>

Strategy: Multi-Judge (Chair Judge)

<BASE_JUDGMENTS> <JUDGE_1> ... </JUDGE_1> <JUDGE_2> ... </JUDGE_2> <JUDGE_3> ... </JUDGE_3> </BASE_JUDGMENTS>
<INSTRUCTIONS> You are the Chair Judge. You have received evaluations from 3 other judges. Synthesize their insights. If they disagree, look at the Ground Truth to break the tie. Provide a final, consolidated evaluation in <THINKING> and scores in <json>. </INSTRUCTIONS>

Strategy: Overseer Audit

<JUDGE_OUTPUT> {judge_output} </JUDGE_OUTPUT>
<INSTRUCTIONS> You are an overseer auditing the judge output. Check ONLY: (i) rubric adherence (is the chosen 0-7 defensible given the rubric and the work?), (ii) faithfulness (did the judge correctly describe what <MODEL_OUTPUT> says vs hallucinate?), (iii) formatting (exactly one <json> block with { "score": int } and nothing after it).
Output ONLY: <json> { "rubric_mismatch_verdict": true/false, "misread_or_hallucination_verdict": true/false, "format_verdict": true/false, "overall_error": true/false, "notes": "one short sentence" } </json> </INSTRUCTIONS>